

Riding on the Fast Track to Debt Risk? Subway, Infrastructure, and the Financing Costs of Local Governments in China^{*}

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April 30, 2026

Abstract

Does debt-financed infrastructure investment pay for itself or lead to fiscal risk? This paper investigates the impact of public infrastructure investment on local government fiscal risks and financing costs. Exploiting the staggered rollout of subway approvals across Chinese cities, we find that subway projects significantly increase the yields of municipal corporate bonds (MCBs) by 23 basis points. The rise in financing costs is concentrated in cities with low subway operational efficiency and is magnified in cities where local politicians face stronger career incentives. We present evidence consistent with two channels: a deterioration in long-term solvency due to massive debt accumulation without commensurate fiscal or land revenue gains, and short-term liquidity risk stemming from a maturity mismatch between long-term assets and short-term debt.

Keywords: Infrastructure Investment; Local Government Debt; Municipal Bond; Fiscal Sustainability; China

JEL Codes: H54, H74, G12, E62, R53

^{*}We thank xxx and seminar participants at xxx for their helpful suggestions.

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1 Introduction

Governments worldwide—ranging from the Biden administration’s Infrastructure Investment and Jobs Act in the US to India’s National Infrastructure Pipeline—are heavily investing in infrastructure. Such investment is widely recognized as a crucial driver of economic development (Glaeser and Poterba, 2020; Donaldson, 2025). A rich body of literature has documented the benefits of subways and other public transit infrastructure, including reducing pollution, promoting productivity, and fostering agglomeration economies.¹ Less is known, however, about the costs and potential risks of infrastructure investment. Such infrastructure entails massive upfront investment, with governments bearing the costs typically through public debt financing. The rapid expansion of these projects raises concern about fiscal sustainability, especially as the global public debt has reached unprecedented levels (Mitchener and Trebesch, 2023).² Despite its apparent policy importance, empirical evidence about its impacts on debt risks and public financing costs remains scarce.

Understanding these questions is increasingly salient in light of the ongoing debate about the sustainability of debt-financed fiscal expansion among both scholars and policymakers. Whether and how infrastructure affects governments’ financial prospects and debt risks remains an open empirical question due to ambiguous theoretical predictions. While some economists highlight negative effects of public debt and view high debt levels as unsustainable (e.g., Reinhart, Reinhart and Rogoff, 2012; Dynan and Elmendorf, 2025), others argue that the fiscal cost of debt may be zero in a low-interest-rate environment and advocate for expansionary infrastructure investment (e.g., Blanchard, 2019; Furman and Summers, 2020). Ideally, infrastructure projects stimulate growth and governments can recover costs through increased tax revenue and land sale (Gupta, Van Nieuwerburgh and Kontokosta, 2022), thereby stabilizing or even decreasing the debt burden. Yet, debt risks could increase if projects underperform or are poorly managed, translating into higher municipal financing costs, reflected by higher municipal bond yields (Schwert, 2017). In addition, political incentives can induce incumbent officials to be fiscally irresponsible (Yared, 2019), leading to over-investment and debt burden accumulation (Song and Xiong, 2026).

In this paper, we study these questions using the context of subway expansion across Chinese cities. China provides an ideal setting to study the impacts of infrastructure. First,

¹ See Anderson (2014); Heblich, Redding and Sturm (2020); Gu et al. (2021); Gendron-Carrier et al. (2022); Agarwal et al. (2025); Koh, Li and Xu (2025).

² Up to 2024, gross government debt reached to 108.5% of GDP in advanced economies and 69.5% of GDP in developing countries (IMF, 2025).

infrastructure investment plays a cornerstone role in China’s economy, amounting to 16.8% of GDP in 2017. A prime example is the rapid expansion of urban transport infrastructure, specifically the subway system, which accounts for nearly one-third of total municipal public infrastructure investment. The subway length has increased from 570 kilometers in 2008 to 7,736 kilometers across 43 cities in 2021. The financial burden of this costly investment falls on local governments. While a relatively small portion of the funding source directly comes from the local fiscal budget, a major portion relies on debt financing largely through local government financing vehicles (LGFVs). These locally controlled, state-owned enterprises are set up to borrow for public projects by bank loans and bonds, known as municipal corporate bonds (MCBs) (Chen, He and Liu, 2020). This allows us to observe comparable measures of public financing cost using the yield of MCBs.

Second, local governments account for the major share of public debt in China.³ The rapid buildup of local government debt, reaching 80% of GDP by 2024, coupled with emerging defaults, has sparked widespread concern about debt risk and sustainability.⁴ Amid recent shrinkage in tax and land sale revenues, the IMF (2024, pp. 23–34) points out that China’s local governments face “daunting fiscal challenge,” noting that “LGFV debt remains unsustainable.” The prominent role of debt-financed infrastructure projects Bai, Hsieh and Song (2016) has led to concern that low-efficiency in these projects may exacerbate local debt risks (Rogoff and Yang, 2024; Qian, Ru and Xiong, 2025).

One empirical challenge to studying the causal effect of infrastructure is that cities commencing projects are likely correlated with other city-specific factors that also affect fiscal outcomes. Another advantage of our setting is that the rollout of subway construction is determined by the central government. The State Council sets three basic criteria for cities applying for approval for a subway project, including city GDP, population, and fiscal revenue. Local governments are required to submit a detailed construction plan to the National Development and Reform Commission (NDRC), subject to the final approval of the State Council. Our empirical strategy exploits this institutional feature in a staggered difference-in-difference (DID) design, and we explicitly control for these criteria variable in a time-varying fashion. We provide event-study results using the approach by Sun and Abraham (2021) to account for heterogeneous treatment effects. Moreover, to ensure comparability, we augment our identification by using a refined control group of cities that propose subway plans but have not yet obtained approval.

First, we examine the effect of subway approval on local government financing cost,

³ By 2023, gross local government debt is 102 trillion RMB, amounting to 80% of GDP. It accounts for 77% of total public debt.

⁴ See for instance, “China’s local debt crisis is about to get nasty”, in the Economist, May 3, 2023. Also, see “China’s cities struggle under trillions of dollars of debt” in the Wall Street Journal, March 6, 2023.

through the lens of bond market outcomes. This approach builds on the idea that the pricing of government bonds reflects market-based broad assessment of fiscal health and debt risk of the corresponding government (Schwert, 2017), and is widely used in the study of local government and sovereign debt risk (e.g., Born, Müller and Pfeifer, 2020; Butler and Yi, 2022; Goldsmith-Pinkham et al., 2023). To this end, we use data on municipal corporate bonds issued by Chinese LGFVs between 2008 and 2017.

Our main finding is that subway projects increase local government financing cost, indicating an overall increase in debt risk. Following the approval of the subway project, we find a significant increase in MCB issue yields by 23 basis points (bp). This result holds when we flexibly control for the variables for approval criteria and is robust to restricting the control group to cities with an intended subway proposal. The event-study result provides evidence in support of the parallel trends assumption. The dynamic effect shows a marked increase in MCB yields by one percentage point, after six years following the subway approval.

We find considerable heterogeneity across cities concealed by the average positive effect on MCB yields. We exploit an internal efficiency measure of the subway system, ridership intensity, as a proxy of infrastructure efficiency (Gendron-Carrier et al., 2022).⁵ Inefficient and less utilized subways would generate lower productivity gains and impose higher pressure for public finance to subsidize subway operation, without delivering insufficient fiscal returns to cover their cost. We find the increase in MCB yields is drastically larger for cities with lower subway ridership intensity, while cities with high-efficiency subways did not experience any meaningful change in public financing cost. Related to the heterogeneity by subway efficiency, we find that the effect on MCB yields is greater for local politicians facing stronger career incentives using the incentives measure by Wang, Zhang and Zhou (2020). This is in line with the narrative that political incentives lead local politicians to promote over-ambitious and likely low-return infrastructure projects by raising excessive debt, undermining long-term fiscal sustainability (Song and Xiong, 2024; any others?).

Having documented the increased local government financing costs, we exploit city-level public finance data, combined with administrative data on land sale transactions and LGFVs' balance sheet data, to investigate the underlying mechanisms. First, we empirically show that the cost of subway investment is financed by the municipal fiscal budget and local public debt. Subway approval increases municipal infrastructure investment from debt by 68% and that from the fiscal budget by 58%. Accordingly, we find a marked increase in the flow of LGFVs net debt financing. Second, we turn to the impacts on fiscal revenue

⁵ Specifically, ridership intensity (10,000 passenger-km per km) is calculated as yearly ridership divided by the length of the subway route.

to explore whether the gains from the subway can pay for the up-front costs. However, we fail to find any significant increase in tax or land sale revenues. To further test the potential land appreciation channel that local governments, as the monopolistic land sellers, can capture a share of the land value appreciation due to infrastructure projects and induced real estate booms (Gyourko et al., 2022), we construct city-level average land prices using residential land transactions data. We again find no change in overall land prices nor land transaction volumes. Third, we show that subway increases local government debt burden using a set of alternative measures. We find that the amount of outstanding debt rises by 26%. The debt-GDP ratio increases by 7 percentage points. Besides the surge in long-term debt stock, leveraging the detailed LGFVs' balance sheet data on the composition of debt structure, we also find a significant increase in short-term debt repayment pressure.

Fourth, we explore the role of the term structure of local government financing in driving the rise in MCB yields. While the potential returns on transit infrastructure typically materialize over a long horizon, the fiscal costs of construction are immediate, and local governments often rely on short-term financing instruments to fund these projects.⁶ This maturity mismatch leads to rollover risk as they have to repeatedly refinance debt, rendering municipal finances more fragile to short-term liquidity shocks and market volatility (Cole and Kehoe, 2000; He and Xiong, 2012). Theoretically, heightened concern about debt risk should predict a more pronounced yield rise in short-term bonds (Arellano and Ramanarayanan, 2012). We document evidence consistent with this prediction: the impact of subway approvals is concentrated in short-term MCB yields. This differential term structure effect is amplified in cities with heavy ex-ante debt burdens, where the fiscal buffer to absorb liquidity shocks is more constrained. In line with this channel, we also find a large increase in refinancing MCB issuance following subway approvals, indicating rollover burden. Taken together, these results suggest that subway projects increase local government financing costs also by heightening the short-term liquidity risk rather than solely long-term debt solvency risk.

Finally, as a complementary analysis, we consider how the investment-driven fiscal pressure would affect fiscal spending on other public goods (Janas, 2026). We find suggestive evidence of modest crowd-out effects. Subway approvals are associated with a decline in healthcare expenditure, an increase in primary school student-teacher ratios, and a reduction in the allocation of land for public services (net of transportation). Crucially, these effects are mainly driven by cities with lower subway ridership intensity.

Related literature. First, we contribute to the literature on the effects of infrastructure

⁶ The average maturity of MCBs in our sample is 5.18 years, substantially shorter than the average 15-year maturity of municipal bonds in the US.

investment, especially urban transit infrastructure (see review by [Donaldson \(2025\)](#)). A large body of work has documented the social and economic benefits of subways. Empirical work shows that subways reduce pollution ([Gendron-Carrier et al., 2022](#)), alleviate urban congestion ([Anderson, 2014](#); [Gu et al., 2021](#); [Barwick et al., 2024](#)), and stimulate commercial activity ([Zheng et al., 2016](#)). By reducing commuting costs, subways enhance intra-city collaborative innovation and improve household financial outcomes ([Severen, 2023](#); [Agarwal et al., 2025](#); [Koh, Li and Xu, 2025](#)). However, there is much less work on the *cost* and *risk* side of infrastructure than on the benefits side. One notable exception is [Brooks and Liscow \(2023\)](#), who document increasing construction cost trends of the US interstate highway system.

Our key contribution is to exploit the staggered rollout of subway approvals across Chinese cities and provide a comprehensive analysis of whether and how subways affect local public finance soundness and debt risks. To our knowledge, we are the first paper to document a positive causal link between subway investment and local government fiscal risk, measured by the rise in public financing costs.⁷ Importantly, our findings highlight the role of infrastructure efficiency in shaping the impacts on fiscal risks. We also extend the research on the political economy of infrastructure provision.⁸ Our finding on the heterogeneous effect by local politicians' promotion incentives indicates that political motives not only affect the allocation of infrastructure ([Burgess et al., 2015](#); [Lehne, Shapiro and Eynde, 2018](#); [Huet-Vaughn, 2019](#); [Fajgelbaum et al., 2024](#); [Chen et al., 2025](#)), but can also sow the seeds of the future public debt risk.

Second, we contribute to the literature on the effects of local government financing. Existing work has studied the effect of debt-raising capacity and financing cost on fiscal and economic real outcomes ([Adelino, Cunha and Ferreira, 2017](#)). On the implication of indebtedness under financial distress, [Janas \(2026\)](#) shows that highly leveraged US city governments in the Great Depression cut expenditure and public goods, leading to development stagnation. A body of work has focused on understanding the relationship between *public* debt and the *private* sector, including the crowding-out effect of public debt on private investment ([Bai, Hsieh and Song, 2016](#); [Cong et al., 2019](#); [Demirci, Huang and Sialm, 2019](#); [Chen et al., 2023](#)), and the reverse crowding-in effect of government deleveraging on private firms ([Hu et al., 2025](#)). Our paper, instead, studies the effects of *public* debt on the *public* sector. We highlight that debt-financed public investment,

⁷ On the fiscal return of subways, [Gupta, Van Nieuwerburgh and Kontokosta \(2022\)](#) analyze the local real estate impact of a subway line extension in New York City. They estimate that only 30% of the value gain from the housing price effect is captured through property tax and is insufficient to cover the cost of the subway.

⁸ Under the same context, political science literature shows that Chinese local officials enjoy higher promotion likelihood after subway approvals by stimulating short-run growth ([Lei and Zhou, 2022](#)).

if underperform, can also become a risk source for the local governments themselves. Furthermore, we add to a large strand of literature on determinants of public borrowing costs. Existing work has identified various factors related to fiscal imbalances, such as austerity policy (Born, Müller and Pfeifer, 2020; Anzoategui, 2022), state pension fund losses (Novy-Marx and Rauh, 2012), population aging (Butler and Yi, 2022), and natural conservation spending (Chen et al., 2024).⁹ We add to this literature by providing new evidence on the role of government investment as another important factor.

Third, our findings broadly speak to the macroeconomics literature on the ongoing debate about fiscal sustainability. Some economists argue that large deficits and high debt levels are harmful. They have negative impacts on economic growth, limit countercyclical fiscal space needed to respond to future crises, threaten debt sustainability, and raise the need for fiscal consolidation (Reinhart, Reinhart and Rogoff, 2012; Romer and Romer, 2019; Kose et al., 2022). In contrast, a group of recent literature argues that the fiscal cost of debt may be zero in a low-interest-rate environment and there may be a “free lunch”, thus advocating for a more proactive fiscal policy (Blanchard, 2019; Furman and Summers, 2020; Angeletos, Lian and Wolf, 2024; Mian, Straub and Sufi, 2025)), viewing infrastructure investment as the engine for “growing out of debt.” By leveraging both city-level and bond-level data, we provide micro-level empirical evidence that debt-financed public investment is a double-edged sword and its net effect could exhibit significant heterogeneity within a country, adding more nuance to the macro theoretic literature.

Finally, our findings relate to the recent research on the interplay among infrastructure investment, land market, and local public finances in China. Recent work has studied how local government debt burden shapes government supply-side intervention in land and real estate markets (Chang, Wang and Xiong, 2025), the relation between land sale pricing and borrowing costs (He et al., 2025), and the linkage between housing markets and local government debt risks (Ambrose, Deng and Wu, 2024). One prevailing narrative posits an intertwined *self-reinforcing cycle* as the successful model of China’s rapid urbanization: local governments leverage land-backed debt to finance infrastructure projects, which in turn boosts land values, real estate boom, and economic growth, thereby generating tax and land revenues to repay debts (Gyourko et al., 2022; Xiong, 2025). Despite its policy significance, we have limited rigorous evidence on the fiscal (in)sustainability of this model. Using the case of subway rollout, we show that infrastructure investment can increase local public borrowing costs and debt burdens without generating commensurate revenue gains.

⁹ Other factors influencing government borrowing costs include migration induced by the COVID-19 pandemic (Gustafson et al., 2025), opioid-related deaths (Cornaggia et al., 2022), state policies for distressed municipalities (Gao, Lee and Murphy, 2019), government monitoring by the media (Gao, Lee and Murphy, 2020), and climate change (Painter, 2020; Goldsmith-Pinkham et al., 2023).

Our findings raise *caution* regarding the universal feasibility of this model and highlight the risk of over-investment in low-return projects and liquidity issues. Instead, its implication on fiscal sustainability hinges critically on infrastructure efficiency (Rogoff and Yang, 2024).

2 Background

2.1 Subways in China

China has witnessed a large boom in subway construction over the past two decades, alongside rapid growth in the urban population. Panel A in Figure 1 illustrates the expansion of subways across Chinese cities. There are a total of 44 cities that had received approval from the central government to construct subway systems by the end of 2016.¹⁰ In 2006, the total length of all subways was only 400 kilometers. By the end of 2021, the total operational length reached 7,736 kilometers, with an additional 4,582 kilometers under construction.¹¹ The financial magnitude of this infrastructure investment is equally striking. Panel B in Figure 1 plots the aggregate share of subway investment in total municipal infrastructure investment in China, which has stabilized at around 30% after 2018. In our sample of cities newly obtaining subway approvals between 2008 and 2017, subway investment accounts for 34.16% of total municipal infrastructure investment,¹² equivalent to 11.02% of the city's general budget revenue on average (Appendix Figure A2). Several studies have documented multiple benefits of subways in China, such as alleviating congestion (Gu et al., 2021; Barwick et al., 2024), reducing air pollution (Li et al., 2019), and stimulating commercial activity (Zheng et al., 2016).

A distinct feature of China's subway development is its highly centralized approval process. Unlike in many decentralized economies where local governments autonomously decide on infrastructure projects, the Chinese central government retains final authority over the approval of massive public investment projects like subways. In 2003, the State Council established three conditions that cities should meet to be eligible to apply for subway plans. These prerequisites, known as the "311" criteria, mandate that a city should have: (i) an annual general fiscal revenue exceeding 10 billion RMB; (ii) a GDP surpassing 100 billion RMB; and (iii) an urban population greater than 3 million. Local governments are required to submit a detailed construction plan to the National Development and Reform Commission (NDRC) for review, subject to the final approval of the State Council

¹⁰ One exception is Baotou, which initially received approval in 2016 but had its construction project halted by the central government in late 2017.

¹¹ Data source: *China Urban Construction Statistical Yearbook*

¹² For certain cities, this share surges to as high as 80% to 90% (Appendix Figure A1)

(No. 81 Document, State Council, 2003). While these criteria only serve as a baseline, the ultimate decision regarding whether and when to grant approval is subject to central government discretion, often influenced by other idiosyncratic factors like macroeconomic policy cycles. Crucially for our research design, this centrally mandated subway rollout process helps to mitigate the empirical challenge that cities commencing infrastructure projects are likely correlated with other city-specific factors that also affect fiscal outcomes. In section 4.1, we will formally discuss the potential concerns about endogenous selection issues.

Subways are a costly upfront investment. As is typical for infrastructure investments in China, the financing and construction of subways are carried out by local governments (Bai, Hsieh and Song, 2016; Amstad, Sun and Xiong, 2020). In practice, a relatively small fraction of subway investment is directly funded through the public budget sourced by land sale revenues. The majority is financed through local government debt (see Section 2.2 for details). Moreover, the operation of subways also imposes a fiscal burden. Subway systems are typically managed by municipal state-owned enterprises (SOEs) that rely heavily on government subsidies to cover operating deficits. The average operating revenue-to-cost ratio was 0.78 in 2016, and has further declined to 0.58 in 2024 (China Association of Metros, 2024).

While many local officials view subways as an attractive investment to boost economic development, the rapid and massive expansion of subway projects across cities has sparked widespread concerns about their efficiency and the possibility of exacerbating local fiscal distress and debt risks. Recent media reports also indicate the severity of pressure. For example, the financial distress of subway operators in Kunming led to frequent wage arrears for subway employees.¹³ In 2018, explicitly due to concern about local government debt risks, the State Council revised the subway approval regulation, significantly raising the eligibility thresholds for project applications (No. 52 Document, State Council, 2018).¹⁴ Consequently, the central government effectively froze approvals for new subway planning since 2017 up to now, restricting authorization only to expansions of existing systems..

¹³ See *China News Weekly*, Sep 27, 2023. <https://www.inewsweek.cn/finance/2023-09-27/19983.shtml>

¹⁴ The policy document (No. 52 Document, State Council, 2018) explicitly states the motivation to “strictly prevent municipal governments from adding new local government debt risks due to urban rail transit construction.” Under the stringent new regulation, only cities with more than 30 billion yuan in fiscal revenue and 300 billion yuan in GDP are eligible to apply for new subway projects, tripling the criteria set by the original 2003 regulation.

2.2 Local public finance and infrastructure investment

Local governments in China have long faced restrictions on borrowing. The 1994 Budget Law legally prohibited local governments from running budget deficits and borrowing any debt. These financial constraints were substantially relaxed during the fiscal stimulus program in 2008 (Bai, Hsieh and Song, 2016). Specifically, local governments were allowed to circumvent these legal restrictions by establishing special off-balance-sheet entities, known as Local Government Financing Vehicles (LGFVs), to borrow on their behalf. As a state-owned enterprise with corresponding local government as the only or dominant shareholder, the primary purpose of LGFVs is borrowing off-balance-sheet public debt without technical violation of the Budget Law. Therefore, the repayment capacity of LGFVs is tightly linked to local public finance (Amstad, Sun and Xiong, 2020, p. 223). Local governments usually transfer land and other public assets into LGFVs as collateral, and pledge future fiscal revenues or subsidies as an implicit guarantee.

Infrastructure investment has been crucial for local governments, not only due to the rising demand of rapid urbanization, but also driven by local politicians' career incentives since their promotions are closely linked with local economic outcomes (Li and Zhou, 2005; Xu, 2011; Wang, Zhang and Zhou, 2020). Even after the stimulus program ended, local governments continued to rely on LGFVs to finance public infrastructure projects through bank loans, trusts, and bonds, namely, municipal corporate bonds (MCBs) (Chen, He and Liu, 2020). In 2015, the central government amended the Budget Law to allow local governments to issue official budgetary bonds. Unlike LGFV debt, the budgetary local government bond is tightly regulated by the central government and its issuance amount is limited by the debt ceiling annually set by the central government.¹⁵

Another important funding source for infrastructure is China's "land finance" system, characterized by public land ownership institution and local governments as the monopoly seller over the land market (Gyourko et al., 2022). Without property taxes, land sale revenues are the most important revenue source controlled by local governments, amounting to 60% 80% of all other tax incomes accrued by city governments. Most of the land revenues are spent on financing public investment and providing firm subsidies. Local government debt financing is intertwined with land finance in two ways: local governments transfer land assets to LGFVs as collateral and use land sale revenues as the source of debt repayment. This creates a self-reinforcing cycle (e.g., Zhou, 2012; Gyourko et al., 2022; Xiong, 2025), widely viewed as the model of China's rapid urbanization. In this cycle,

¹⁵ Concurrently, the Ministry of Finance launched a "debt swap program" during 2015 to 2018, under which local governments can refinance a certain portion of their maturing LGFV debts by issuing lower-interest official local government bonds

debt-financed infrastructure investments—such as subways—enhance economic growth potential, improve urban amenities, and stimulate the real estate sector, which would in turn expand the future tax base and drive-up land values, translating to higher land sale revenues. Local governments can then capture the social and fiscal returns to service existing debts and fund subsequent projects. Ideally, this model allows city governments to achieve rapid urbanization while maintaining debt sustainability, provided that the economic efficiency of the underlying investments remains sufficiently high.

Notwithstanding several tightening policy attempts to limit LGFV debt accumulation, local government debt continued to surge to a record high level during the past two decades. As illustrated in Figure 2, gross local government debt—comprising both official budgetary bonds and LGFV debt—has reached approximately 80% of GDP by 2023. The IMF (2024) projects that China’s augmented public debt would rise to 165% of GDP by 2033. This rapid debt buildup, and recent LGFV debt defaults have led to heightened concerns and policy debates regarding local fiscal sustainability and debt risks.¹⁶ Considering the potential heightened fiscal and debt strains, the central government even directly restricted public investment projects by twelve highly indebted provinces in 2024.¹⁷ Despite the large amount of investment and significant policy importance, there has been little empirical evidence on how these infrastructure projects like subways affect local fiscal risks.

2.3 Municipal corporate bonds

Our empirical analysis focused on Municipal corporate bonds (MCBs), which refer to bonds issued by LGFVs.¹⁸ The word “corporate” merely indicates that they are corporate bonds in a legal sense since LGFVs nominally are corporations. However, they are widely treated as quasi-municipal bonds by both market participants and financial regulation, with implicit guarantees from corresponding local governments. They are functionally similar to the municipal bonds issued by local governments in the US and other countries. Driven by the rollover pressures of the 2008 stimulus bank loans and the continued need for local government financing, the MCB market has expanded quickly, catalyzing the development of China’s corporate bond markets (Chen, He and Liu, 2020).

Because local governments ultimately back these bonds issued by LGFVs on their behalf, MCB yields reflect local government borrowing costs and can serve as the market-based barometers of local fiscal risk (He and Wei, 2023). In general, MCB yields capture investors’

¹⁶ See Caixin Global, May 22, 2023, “China’s Effort to Move Mountain of ‘Hidden Debt’ Faces Uphill Climb.”

¹⁷ See Reuters, Jan 19, 2024, “Exclusive: China orders indebted local governments to halt some infrastructure projects.”

¹⁸ It is “cheng-tou-zhai” in Chinese, which literally translates to “city investment bonds.”

pricing of a local government’s underlying fiscal stance and credit risk, with the financing costs varying significantly across cities (Appendix Figure [appfig_line_monthmean_MCB_rate](#)). Indeed, an extensive volume of sell-side research and credit rating reports in China focus on local governments’ fiscal condition and debt burden to forecast MCBs yield dynamics. This motivates our study to use MCB yields as a proxy of local fiscal risks, net of time fixed effects to hold the general interest rate environment.

3 Data

We combine multiple sources of bond-level data, city-level economic and public finance data, and individual land transaction data across Chinese cities from 2008 to 2017.¹⁹ Below, we describe the source and construction of our dataset.

Subways. We begin by collecting information on subway plan approvals. We first draw the city list of new subway approvals every year from the *Annual Statistics and Analysis Report of China Association of Metros*, and then we identify the approved subway plan for each city from the website of the NDRC. Appendix Table [A2](#) reports the full list of cities and respective approved years. To measure the operational efficiency of subways, we also collect ridership data from the *Annual Report of China Association of Metros*. Specifically, we employ a measure of ridership intensity (10,000 passenger-km per km), calculated as the total annual ridership divided by the length of the subway route.

Sample definition. Our sample includes 30 cities as the treatment group that received subway plan approvals between 2008 and 2017, and 256 cities serving as the never-treated group. Following standard practices in staggered difference-in-differences (DID) designs, we exclude 14 cities whose subway plans were already approved prior to 2008.

Moreover, to improve comparability across cities, we manually collect information on cities that have subway construction proposals but have not yet obtained official approvals. Prior to submitting subway applications, cities are required to conduct feasibility studies, formulate route plans, complete environmental impact assessments, establish subway construction leading groups comprising relevant municipal officials, and set up state-owned firms explicitly named for subway construction. Most of this information is publicly disclosed and widely reported by the media. We manually conduct city-by-city searches on the Internet and identify 122 cities with intended subway plans. In most of our empirical analyses, we use this subset of cities as a refined control group.

Municipal corporate bond (MCB) data. We collect data on municipal corporate bonds

¹⁹ We end our sample period in 2017 to avoid potential confounding factors arising from a major shift in subway approval policies enacted that year (see Section 2.1).

(MCBs) from Wind Information Co. (WIND), the leading financial data vendor in China. Matched with our sample cities, the dataset comprises 8,491 MCBs issued by 1,774 LGFVs.²⁰ For each MCB issuance, the data provide various bond characteristics, including actual issuing amount, issuing date, maturity date, issuer's location, bond rating, and issuer's rating. The data also include the issuance purpose for each MCB, which we broadly classify into two categories: (a) debt refinancing and (b) operations and investment. Appendix Table A1 presents the summary statistics for the MCBs. The average yield is 6.07%, and there is substantial variation in MCB yields, with the 5th percentile at 3.64% and the 95th percentile reaching as high as 8.30%. The average maturity is 5.18 years.

Local government debt data. To measure local government debt burden, we follow the existing literature to use the interest-bearing debt of LGFVs (Huang, Pagano and Panizza, 2020), which encompasses bonds, bank loans, trust loans, and other non-standard financing instruments. We collect these data from LGFVs' balance sheets via WIND and aggregate the measure to the city-year level. Beyond the *stock* of outstanding debt, leveraging detailed balance sheet data on debt composition, we can measure the *flow* of debt repayment pressure using the amount of LGFV debt maturing in the next year.

We complement these LGFV debt data with official local government bonds, which only emerged after 2015. Since there is no one-step repository for official local government bonds at the city level, we manually collect these data from budget and final account reports published on prefecture and provincial government websites. Combining official local government bonds with LGFV debt provides a more comprehensive measure of the overall local government debt burden.

Land parcel transaction data. We obtain the universe of land parcel transactions from the Ministry of Natural Resources of China.²¹ For each transaction, this database provides the transaction price, acreage, land grade, geographic location, transaction method, and designated use type. In our analysis of the impact on land prices, we restrict our sample to 233,246 residential land transactions conducted through public tender, auction, or listing, with market-based prices.²² We geocode the precise latitude and longitude coordinates of each parcel based on its address.

Other city-level economic and fiscal data. Finally, we collect city-level data on a set of socioeconomic variables from the *China City Statistical Yearbook*, the *China Urban Construction Statistical Yearbook*, and the 2010 Population Census. We also obtain data on

²⁰ We exclude MCBs in the form of convertible bonds, exchangeable bonds, and asset-backed securities, due to their non-standard structure, limited market size.

²¹ The data is collected from the China Land Market website (<http://www.landchina.com>)

²² We exclude other land use types, such as industrial land, because these parcels are frequently sold at extremely low or near-zero prices and thus play a minimal role in generating local government land sale revenues (He et al., 2025).

local fiscal revenues and expenditures from city-level government budget and final account reports.

4 Subway approval and MCB yield

4.1 Empirical strategy

Our primary goal is to study the effects of subway approvals on local public financing costs. Our baseline analysis uses the bond-level data and exploits the staggered rollout of subway approvals across cities in a difference-in-differences (DID) design, comparing changes in MCB yields issued in cities with approved subway plans against those in the control group. We estimate the following specification:

$$Y_{ict} = \beta \text{Subway}_{ct} + \delta Z_i + \gamma X_{ct} + \lambda_c + \theta_t + \epsilon_{ict} \quad (1)$$

where Y_{ict} is the outcome of interest for bond i , issued by city c in year t . The main explanatory variable, Subway_{ct} is a dummy variable that equals one if city c had received the approval of the subway plan by the central government at year t , and zero otherwise. Z_i denotes a vector of bond characteristics, including logged issue size, time to maturity, indicator for guarantee, bond rating, and issuer rating. We include city fixed effects λ_c to capture all time-invariant confounding factors. θ_t are year fixed effects capturing the common shocks like overall monetary policy and national interest environment. We cluster standard errors at the city level. β is the coefficient of our primary interest, capturing the effect of subway approvals.

The key assumption for causal identification is that outcomes of bonds issued by treated and control cities would have evolved along parallel trends, in the absence of subway planning approvals. The primary identification concern is that subway approvals are not randomly assigned across cities. Table 1 compares a set of initial city characteristics of the control and treatment group, revealing that they indeed exhibit significant differences (column 4). We address this issue in several ways. First, we improve the comparability of our sample by restricting the control group to a subgroup of cities that have proposed subway construction plans but have not yet obtained approvals. Using this refined control group ensures that our estimates account for the endogenous intention to apply for subway construction.

Second, we leverage the institutional features of the centralized review process for subway projects and explicitly control for treatment assignment criteria set by the central government introduced in Section 2.1. Specifically, we include the 2008 baseline levels

of city GDP, population, and fiscal revenues, interacted with year fixed effects, in the vector of controls X_{ct} . To better understand the determinants of subway approvals, we estimate various Cox hazard models to examine the correlation between city characteristics and approvals. As shown in Appendix Table A3, except for fiscal revenue, none of the other economic and public finance variables can significantly predict the timing of subway approval. In support of this approach, column 5 of Table 1 shows that, conditional on controlling for baseline fiscal revenue, the treated and control cities are highly balanced along a wide range of characteristics, including GDP, GDP per capita, demographic features, and public debt burden.

Third, we formally check the validity of the parallel trends assumption by estimating an event-study specification to track the dynamic treatment effects relative to the year just preceding the approvals:

$$Y_{ict} = \sum_d \beta_d \text{Subway}_{ct} T_{d(ct)} + \delta Z_i + \gamma X_{ct} + \lambda_c + \theta_t + \epsilon_{ict} \quad (2)$$

where $T_{d(ct)}$ are indicators for the periods relative to the treatment year. To account for the econometric issues of negative weighting and heterogeneous treatment effects inherent in standard two-way fixed effects (TWFE) staggered DID designs, we estimate the event-study specifications throughout this paper using the robust estimator proposed by [Sun and Abraham \(2021\)](#). If the parallel trends assumption is satisfied, we should expect that the coefficients β_d corresponding to pre-treatment periods are close to zero.

4.2 Main results

Table 2 presents our main results of the impact of subway approvals on municipal corporate bond (MCB) yields. Column 1 reports the baseline specification (equation 1) including city and year fixed effects along with bond-level controls. On average, subway approvals are associated with a 23 basis point (bp) increase in MCB yields. This translates to an increase of 3.8% in local public financing costs. In column 2, we further include baseline criteria variables related to subway approvals interacted with year fixed effects: city GDP, population, and fiscal revenues in 2008. Reassuringly, the estimate remains virtually unchanged after the inclusion of these treatment assignment controls. To further mitigate concerns regarding the potential endogenous selection into subway planning, Column 3 restricts the control group sample to cities that have proposed subway plans but have not yet obtained approval. The estimated coefficient remains similar in both magnitude and significance, suggesting that our results are unlikely to be confounded by the endogenous intention to build subways or initial city differences relevant to subway approval rollout.

Figure 3 displays the event-study results along with the 95% confidence intervals from estimating equation 2. Results are estimated following the approach by Sun and Abraham (2021) to account for heterogeneous treatment effects in staggered DID estimation. We observe little difference between the trends of treated and control cities prior to the rollout of subway approvals, lending support to the validity of the parallel trends assumption. After the approvals of the subway project, there is a marked and persistent increase in MCB yields for subway cities relative to the control group. The gap widens gradually to around one percentage point six years after the subway approval.

Robustness checks. We conduct a series of robustness checks. The detailed results are reported in the Appendix B. First, we re-estimate our baseline specification (equation 1) on the average treatment effect using the interaction-weighted estimator proposed by Sun and Abraham (2021). The results remain similar (Appendix Table B1). Second, our results are robust to using alternative measures of local public financing costs. Our result holds when we use the credit spread calculated as the difference in yields between MCB and treasury bonds with the equivalent maturity (Appendix Table B2). Third, we also use the secondary market transactional yield and spread instead of the primary issuance yield. In both cases, the estimated coefficients remain positive and significant (Appendix Table B3).

Finally, we consider whether our results are confounded by the debt swap program after the new Budget Law in 2015. Between 2015 and 2018, the central government allowed provincial governments to issue approximately 12.2 trillion RMB official local government refinance bonds to swap out existing high-interest LGFV debt. This program alleviated liquidity pressure and reduced debt servicing burdens for LGFVs. To account for this, we construct a measure of debt swap intensity as an additional control variable, calculated as the aggregated issuance of special refinance government bonds scaled by the 2014 LGFV outstanding debt at the province-year level.²³ Our main findings remain unchanged (Appendix Table B4).

4.3 Heterogeneity: operational efficiency and career incentives of local politicians

Next, we investigate the heterogeneous effect of subway approval on public financing cost. We begin by estimating a modified version of equation (1) where we allow the effect to vary in each city. Figure 4 shows the resulting city-specific heterogeneous effects on MCB yield, with each bar indicating an estimated coefficient for a city. We find considerable heterogeneity across cities concealed by the average positive effect on MCB yields. While most cities show increased MCB yield, the effects are modest for some cities. Moreover,

²³ Data on special government refinance bonds are collected from China Electronic Local Government Bond Market Access (<https://www.celma.org.cn/>).

three cities appear to experience a decrease in public financing costs.

First, we explore the heterogeneity driven by the operational efficiency of the subways. The underlying hypothesis is that highly efficient subway investments can generate sufficient economic returns and positive fiscal spillovers to cover their debt financing costs. They may not increase local fiscal risks and even reduce long-term debt burden by sustainably expanding the local tax base. Conversely, inefficient and underutilized investments are expected to significantly raise local borrowing costs. To test this, we utilize subway ridership intensity as a proxy for operational efficiency. We classify treated cities into “high” and “low” ridership groups based on whether their ridership intensity was above or below the sample median. Columns 1 and 2 of Table 3 reports the results of interacting the treatment indicator with these two group dummies. We find that the positive impact of subway approvals on MCB yields is mainly driven by cities with low-efficiency subways, exhibiting an increase of 33.7 to 34.5 basis points. In stark contrast, for cities with high ridership, the estimated coefficients are small and statistically indistinguishable from zero.

Then we turn to the political economy dimension: the returns and utilization of infrastructure projects are also shaped by local politicians. Massive infrastructure investment is attractive to them to boost growth, as local leaders’ promotion is closely related with local economic outcomes in China (Li and Zhou, 2005; Xu, 2011; Jia, Kudamatsu and Seim, 2015). Strong incentives can induce short-termism, leading local officials to pursue overly ambitious infrastructure plans and accumulate high leverage (Song and Xiong, 2026). Following Wang, Zhang and Zhou (2020), we measure politicians’ career incentives by the rule-derived promotion likelihood based on a combination of their age of tenure and bureaucratic rank. As shown in Columns 3 and 4 of Table 3, we find that the effect of subway approvals on government borrowing costs is much larger if local politicians face stronger career incentives.²⁴ These results suggest that political incentives of local leaders exacerbate the fiscal risks associated with debt-financed infrastructure expansions.²⁵

5 Mechanisms

The empirical findings so far have established that subway approvals have a positive effect on city government financing cost, indicating increased local fiscal risk priced in by the market. In this section, we investigate the underlying channels through which subway projects affect MCB yields. We present evidence in support of two channels: debt solvency

²⁴ We partition the sample based on the median career incentive of the city’s party secretary at the time the subway was approved.

²⁵ This finding complements Wang, Zhang and Zhou (2020) that career incentives of city leaders lead to outward urban expansion beyond what would be justified by their cities’ population growth.

risk and short-term liquidity risk.

5.1 Debt and fiscal cost of infrastructure investment

We first empirically demonstrate that the sheer scale of subway investments places substantial pressure on both municipal fiscal budgets and local public debt. Table 4 presents the results relating subway approvals to infrastructure investment outcomes using city-year level panel data. The approval of subway construction leads to a substantial 41% increase in municipal infrastructure investment (column 1), highlighting the outsized role of subways in local capital expenditures. When breaking down the investment by funding sources, we find a 68% increase in debt-financed investment (column 2). Meanwhile, investment funded directly by the fiscal budget also experiences a 58% surge (column 3). Given that the central government budget accounts for only 1% of the total municipal infrastructure investment, this increase is driven by the local fiscal budget. In column 4, we further look at the net increase in LGFV debt. Consistent with the debt financing nature of subway investment, we find a significant increase in the flow of LGFVs net debt financing, equivalent to a 100% increase relative to the sample mean.

5.2 Impacts on fiscal revenue

Next, we turn to the revenue side to investigate whether subway investments generate sufficient fiscal returns to cover their substantial up-front costs and the induced debt burden. In theory, local governments can capture the returns on public infrastructure through two primary channels. First, highly efficient subway systems can significantly improve urban amenities, attract population inflows, and boost local productivity, thereby expanding the future general tax base. Second, through a land value capture mechanism closely linked to the real estate sector, subway construction can lead to an instant appreciation in surrounding land prices, thereby boosting land sale revenues. This is similar to the housing value appreciation channel captured by the property tax in the US (Gupta, Van Nieuwerburgh and Kontokosta, 2022). Moreover, since land sale revenue is also an important funding source for infrastructure in China, a surge in such revenues could decrease the need for external debt financing, thereby mitigating fiscal risks.

In Table 5, we empirically examine these revenue-side channels. Strikingly, we find no evidence that subway approvals generate meaningful direct fiscal returns. Column 1 shows that the estimated effect on log tax revenues is small and statistically indistinguishable from zero. In Column 2, while we observe a modest positive coefficient for land sale revenues, the estimate is also insignificant. Columns 3 and 4 report the impacts on the land sale area

and average land prices, respectively. Neither volume nor price exhibits any significant change following the subway approval. Event-study estimates further corroborate these null effects, showing no discernible pre-trends or treatment effects (Appendix Figure A4).²⁶

Discussion on land price effect. To further unpack the muted response in land price, we use the micro-level residential land transactions data. Appendix Table A5 reports the parcel-level estimates. Consistent with our city-level result, the average treatment effect on land prices remains small and insignificant (Columns 1–3). We investigate the heterogeneous effects within cities by distance to subway station. In Appendix Figure A5, we estimate the heterogeneous price effects based on the distance. Compared to control cities, parcels located within 4 kilometers of a station experience massive price premiums by more than 80%, while those between 4 and 7 kilometers see a moderate 25% increase. However, the price effect actually turns negative (-17%) for land transactions located beyond 7 kilometers, which account for 66% of the total residential land sale revenues.

The decline in residential land prices in peripheral areas might be driven by spatial relative diversion of public goods and amenities. Given the massive costs of subway construction, financially constrained local governments may reallocate limited public resources away from the periphery to support subway-adjacent developments. Appendix Figure A6 provides descriptive evidence in line with this. Examining the spatial distribution of newly supplied land zoned for public services as a proxy for public goods and amenities, we find a relative decline in the share of new public good land in treated cities beyond 7 kms from any subway station, with a corresponding surge in public good land concentrated within 2 kms of the stations.²⁷

Taken together, these empirical patterns suggest that subway approvals induce a *spatial reallocation* of land demand within the city toward station-adjacent areas, rather than generating an aggregate real-estate boom. In line with this interpretation, we observe no change in aggregate city-level real estate investment following subway approvals (Appendix Figure A7). These results can explain the aggregate null effect on land sale revenue.

5.3 Impacts on local public debt burden

We proceed to show that subway projects lead to local government debt burden accumulation using a set of alternative measures. Table 6 reports the results. Using the interest-bearing debt of LGFVs aggregated at the city level, we find that subway approvals

²⁶ As a robustness check, we re-estimate the models using per capita measures for fiscal and land revenues. The results remain consistent with the null findings (Appendix Table A4)

²⁷ These land parcels are land zoned for public facility, education, health, green land, and civil infrastructure. We exclude transportation infrastructure to avoid mechanical effects of subway construction.

lead to a 26% increase in the overall debt stock (columns 1-2). To evaluate the scale of this debt relative to the local GDP, columns 3-4 show that the debt-to-GDP ratio increases by 7 percentage points. Besides the surge in outstanding long-term debt stock, leveraging the detailed LGFVs' balance sheet data on debt maturity structures, we also find a 3.2 percentage points increase in short-term debt repayment pressure, defined as the amount of LGFV debt maturing in the subsequent year divided by city GDP. Figure 5 plots the corresponding event-study results following the approach by Sun and Abraham (2021). All three measures show a significant increase after subway approvals with no discernable pre-trends. We also additionally control for the debt swap policy that provincial governments issue official local government refinance bonds to swap out part of existing LGFV debt in columns 2, 4, and 6. Reassuringly, our findings remain virtually unchanged.

Our results are robust to a variety of alternative debt measures. As reported in Appendix Table A6 and Figure A8, the findings remain similar when we use the gross local government debt-to-GDP ratio, which incorporates official budgetary local government bonds issued post-2015. Moreover, the results hold when we measure the debt burden relative to the city's total fiscal budget, defined as the sum of general public budget revenues, fiscal transfers, and land sale revenues.

5.4 Term structure and rollover risk

In this section, we explore another mechanism underlying the increase in public financing costs by focusing on the role of the term structure of local government financing. A fundamental friction in municipal infrastructure financing is the maturity mismatch: while the potential economic and fiscal returns of infrastructure typically materialize over a long time, the massive fiscal costs of construction are immediate. Chinese local governments predominantly rely on relatively short-term financing instruments to fund these long-term projects. As illustrated in Figure 6, the average maturity of MCBs in our sample is merely 5.18 years and exhibits a further downward trend over our sample period, which is substantially shorter than the average 15-year maturity typical of the US municipal bond market.

This maturity mismatch necessitates that local governments have to repeatedly refinance debt, leading to rollover risk that renders municipal finances more fragile to short-term liquidity shocks and financial market volatility (He and Xiong, 2012). In extreme cases, excessive reliance on short-term debt can open the door to self-fulfilling liquidity crises (Cole and Kehoe, 2000). The general rollover pressure of local governments in our context is evident, as shown by the dashed line in Figure 6. The fraction of newly issued MCBs designated for refinancing purposes rapidly increased to nearly 80% by 2017.

Theoretically, heightened concern about debt risk should predict that short-term yields rise more, as default is more likely in the near future ([Arellano and Ramanarayanan, 2012](#)). We examine this in Table 7 by splitting the bond sample based on maturity. Consistent with the prediction, columns 1 and 2 show that the impact of subway approvals is concentrated in short-term MCB yields, which exhibit an increase of 34 basis points, whereas long-term bond yields show no significant change. Moreover, columns 3 and 4 show that this heterogeneous effect by bond maturity is more pronounced in cities with high initial debt burdens (measured by the LGFV debt-to-GDP ratio in 2008). This is consistent with the notion that these highly indebted cities have a lower capacity to absorb additional short-term liquidity shocks, prompting the market to price in a substantial rollover risk premium.

We also provide additional corroborating evidence on the rollover behaviors. In Table 8, we utilize the designated usage of the MCB issuance to categorize bonds into refinancing bonds and operating/investment bonds. Consistent with the heightened rollover pressure, we find that subway approvals are associated with a surge in MCB issuances for debt refinancing, measured by the probability (column 1) and the absolute amount (column 2). In contrast, MCB issuances for operations or investments show no significant change (columns 3 and 4).

Taken together, these empirical patterns we have shown in the Section 5 suggest that both long-term solvency concern and short-term liquidity risk are at play driving the rise in public financing costs. As another robustness check, our city-year level results are robust to alternative estimation using the synthetic difference-in-difference estimator by [Arkhangelsky et al. \(2021\)](#), which generates matched controls for the affected municipalities and relaxes the parallel trends assumption (see Table A7). Our results remain the same following the staggered DID estimators by [Callaway and Sant’Anna \(2021\)](#) (see Table A8).

5.5 Impacts on local public goods

Finally, as a complementary analysis, we provide suggestive evidence on the real consequences of the debt pressure induced by subway construction. Specifically, we consider whether the debt-driven fiscal pressure would force local governments to cut other fiscal spending, thereby crowding out the provision of other public goods ([Janas, 2026](#)).

Table 9 presents the empirical results. Panel A provides evidence of a modest crowding-out effect on average. Following a subway approval, local public healthcare expenditure declines by 9.1% (Column 1). Furthermore, we observe an increase in the primary school student-teacher ratio (Column 2), indicating a deterioration in public education quality. In Column 3, we observe a reduction in the land area share allocated to public goods (net

of transportation), though this average effect is statistically insignificant. Crucially, Panel B reveals that these crowding-out effects are mainly driven by cities with low subway efficiency. In contrast, cities with highly utilized subways experience no such crowding-out effects.

6 Conclusions

Against the backdrop of a global surge in public debt and its policy debate, this paper provides novel micro-level evidence on the impact of infrastructure investment on public financing cost. Exploiting the staggered rollout of subway approvals across Chinese cities in a difference-in-differences framework, we find that the approval of subway increases local government financing cost, indicating an overall increase in debt risk. Following the approval of subway project, we find a significant increase in MCB issue yields by 23 basis points.

The average positive effect on MCB yields conceals considerable heterogeneity. We find that the increase in public financing costs is drastically larger for cities with lower operational efficiency of subway, and for local politicians facing stronger career incentives. This indicates that political motives lead officials to promote over-ambitious, low-return projects that raise excessive debt. Investigating the underlying mechanisms, we empirically show that subway investments are heavily financed by municipal budgets and local public debt, leading to a marked increase in the flow of LGFVs' net debt financing. However, we fail to find any significant increase in tax revenues, land sale revenues, or overall land prices. Consequently, subway approvals significantly increase the local government debt burden, reflected by a 26% rise in outstanding debt, a 7-percentage-point increase in the debt-to-GDP ratio, and a substantial surge in short-term debt repayment pressure.

Our findings can shed lights on policy implications and the broader macroeconomic debate on fiscal sustainability. Specifically, our empirical evidence cautions against the prevailing narrative in China that debt-financed infrastructure consistently pays for itself through subsequent land revenues return and economic growth. Our heterogeneity analysis based on subway ridership suggests a nuanced reality: while high-performance infrastructure indeed may not necessarily compromise local fiscal health, infrastructure projects lacking sufficient operational efficiency—coupled with imprudent debt management—can precipitate both long-term structural debt overhangs and local liquidity stress.

References

- Adelino, Manuel, Igor Cunha, and Miguel A. Ferreira.** 2017. "The economic effects of public financing: Evidence from municipal bond ratings recalibration." *The Review of Financial Studies*, 30(9): 3223–3268.
- Agarwal, Sumit, Yeow Hwee Chua, Pulak Ghosh, Soumya Kanti Ghosh, and Li-uyang She.** 2025. "Mortgages, subways and automobiles." *Journal of Public Economics*, 252: 105522.
- Ambrose, Brent W., Yongheng Deng, and Jing Wu.** 2024. "Understanding the risk of China's local government debts and its linkage with property markets." Working paper.
- Amstad, Marlene, Guofeng Sun, and Wei Xiong.** 2020. *The handbook of China's financial system*. Princeton University Press.
- Anderson, Michael L.** 2014. "Subways, strikes, and slowdowns: The impacts of public transit on traffic congestion." *American Economic Review*, 104(9): 2763–2796.
- Angeletos, George-Marios, Chen Lian, and Christian K. Wolf.** 2024. "Can deficits finance themselves?" *Econometrica*, 92(5): 1351–1390.
- Anzoategui, Diego.** 2022. "Sovereign spreads and the effects of fiscal austerity." *Journal of International Economics*, 139: 103679.
- Arellano, Cristina, and Ananth Ramanarayanan.** 2012. "Default and the maturity structure in sovereign bonds." *Journal of Political Economy*, 120(2): 187–232.
- Arkhangelsky, Dmitry, Susan Athey, David A Hirshberg, Guido W Imbens, and Stefan Wager.** 2021. "Synthetic Difference-in-Differences." *The American Economic Review*, 111(12): 4088–4118.
- Bai, Chong-En, Chang-Tai Hsieh, and Zheng Song.** 2016. "The long shadow of China's fiscal expansion." *Brookings Papers on Economic Activity*, 2016(2): 129–165.
- Barwick, Panle Jia, Shanjun Li, Andrew Waxman, Jing Wu, and Tianli Xia.** 2024. "Efficiency and equity impacts of urban transportation policies with equilibrium sorting." *American Economic Review*, 114(10): 3161–3205.
- Blanchard, Olivier.** 2019. "Public debt and low interest rates." *American Economic Review*, 109(4): 1197–1229.
- Born, Benjamin, Gernot J. Müller, and Johannes Pfeifer.** 2020. "Does austerity pay off?" *The Review of Economics and Statistics*, 102(2): 323–338.
- Brooks, Leah, and Zachary Liscow.** 2023. "Infrastructure costs." *American Economic Journal: Applied Economics*, 15(2): 1–30.

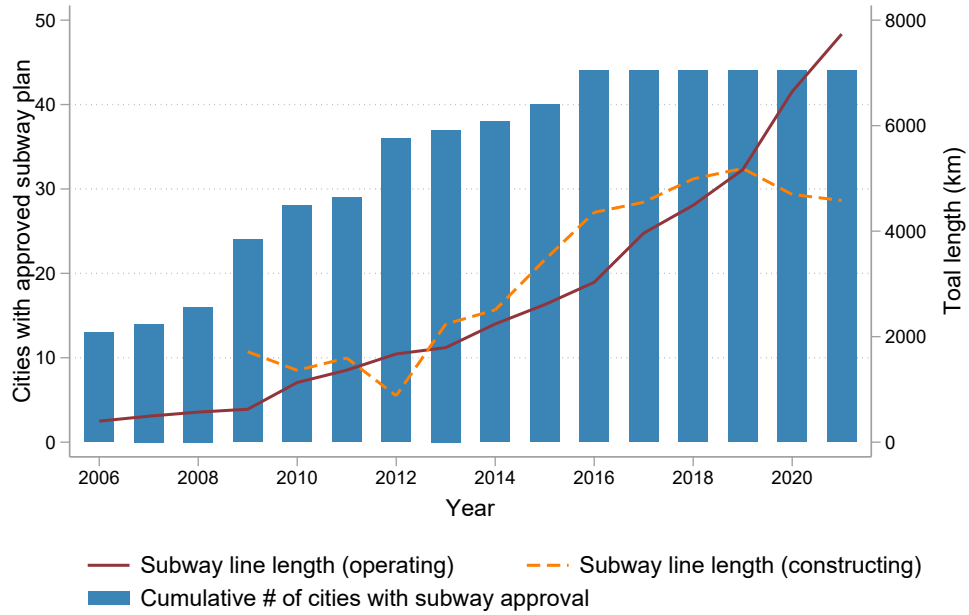
- Burgess, Robin, Remi Jedwab, Edward Miguel, Ameet Morjaria, and Gerard Padró i Miquel.** 2015. "The value of democracy: Evidence from road building in Kenya." *American Economic Review*, 105(6): 1817–1851.
- Butler, Alexander W., and Hanyi Yi.** 2022. "Aging and public financing costs: Evidence from U.S. municipal bond markets." *Journal of Public Economics*, 211: 104665.
- Callaway, Brantly, and Pedro HC Sant'Anna.** 2021. "Difference-in-Differences with multiple time periods." *Journal of Econometrics*, 225(2): 200–230.
- Chang, Jinfan, Yuheng Wang, and Wei Xiong.** 2025. "Price and volume divergence in China's real estate markets: The role of local governments." *The Review of Financial Studies*, 39(2): 343–386.
- Chen, Fukang, Minhao Gao, Lin William Cong, and Jacopo Ponticelli.** 2024. "Pricing the priceless: The financial cost of biodiversity conservation." NBER Working Paper.
- Chen, Kaiji, Haoyu Gao, Patrick C. Higgins, Daniel F. Waggoner, and Tao Zha.** 2023. "Monetary stimulus amidst the infrastructure investment spree: Evidence from China's loan-level data." *The Journal of Finance*, 78(2): 1147–1204.
- Chen, Zhuo, Zhiguo He, and Chun Liu.** 2020. "The financing of local government in China: Stimulus loan wanes and shadow banking waxes." *Journal of Financial Economics*, 137(1): 42–71.
- Chen, Ziyang, Ting Chen, Yatang Lin, and Jin Wang.** 2025. "Building tall, falling short: An empirical assessment of Chinese skyscrapers." *Journal of Urban Economics*, 145: 103731.
- China Association of Metros.** 2024. "Annual Statistics and Analysis Report of Urban Rail Transit." China Association of Metros.
- Cole, Harold L., and Timothy J. Kehoe.** 2000. "Self-fulfilling debt crises." *The Review of Economic Studies*, 67(1): 91–116.
- Cong, Lin William, Haoyu Gao, Jacopo Ponticelli, and Xiaoguang Yang.** 2019. "Credit allocation under economic stimulus: Evidence from China." *The Review of Financial Studies*, 32(9): 3412–3460.
- Cornaggia, Kimberly, John Hund, Giang Nguyen, and Zihan Ye.** 2022. "Opioid crisis effects on municipal finance." *The Review of Financial Studies*, 35(4): 2019–2066.
- Demirci, Irem, Jennifer Huang, and Clemens Sialm.** 2019. "Government debt and corporate leverage: International evidence." *Journal of financial economics*, 133(2): 337–356.
- Donaldson, Dave.** 2025. "Transport infrastructure and policy evaluation." NBER Working Paper.
- Dynan, Karen, and Douglas Elmendorf.** 2025. "Putting US fiscal policy on a sustainable path." *Journal of Economic Perspectives*, 39(4): 27–48.

- Fajgelbaum, Pablo D., Cécile Gaubert, Nicole Gorton, Eduardo Morales, and Edouard Schaal.** 2024. "Political preferences and transport infrastructure: Evidence from California's high-speed rail." Working paper.
- Furman, Jason, and Lawrence H. Summers.** 2020. "A reconsideration of fiscal policy in the era of low interest rates." Working paper.
- Gao, Pengjie, Chang Lee, and Dermot Murphy.** 2019. "Municipal borrowing costs and state policies for distressed municipalities." *Journal of Financial Economics*, 132(2): 404–426.
- Gao, Pengjie, Chang Lee, and Dermot Murphy.** 2020. "Financing dies in darkness: The impact of newspaper closures on public finance." *Journal of Financial Economics*, 135(2): 445–467.
- Gendron-Carrier, Nicolas, Matthew González-Navarro, Stefano Polloni, and Matthew A. Turner.** 2022. "Subways and urban air pollution." *American Economic Journal: Applied Economics*, 14(1): 164–196.
- Glaeser, Edward L., and James M. Poterba.** 2020. "Economic analysis and infrastructure investment." NBER Working Paper.
- Goldsmith-Pinkham, Paul, Matthew T. Gustafson, Ryan C. Lewis, and Michael Schwert.** 2023. "Sea-level rise exposure and municipal bond yields." *The Review of Financial Studies*, 36(11): 4588–4635.
- Gupta, Arpit, Stijn Van Nieuwerburgh, and Constantine Kontokosta.** 2022. "Take the Q train: Value capture of public infrastructure projects." *Journal of Urban Economics*, 129: 103422.
- Gustafson, Matthew, Peter Haslag, Daniel Weagley, and Zihan Ye.** 2025. "A flash in the pan(demic)? Migration risks and municipal bonds." *Management Science*.
- Gu, Yizhen, Chang Jiang, Junfu Zhang, and Ben Zou.** 2021. "Subways and road congestion." *American Economic Journal: Applied Economics*, 13(2): 83–115.
- Gyourko, Joseph, Yang Shen, Jing Wu, and Rongjie Zhang.** 2022. "Land finance in China: Analysis and review." *China Economic Review*, 76: 101868.
- Heblich, Stephan, Stephen J. Redding, and Daniel M. Sturm.** 2020. "The making of the modern metropolis: Evidence from London." *The Quarterly Journal of Economics*, 135(4): 2059–2133.
- He, Zhiguo, and Wei Wei.** 2023. "China's financial system and economy: A review." *Annual Review of Economics*, 15: 451–483.
- He, Zhiguo, and Wei Xiong.** 2012. "Rollover risk and credit risk." *The Journal of Finance*, 67(2): 391–430.
- He, Zhiguo, Scott Nelson, Yang Su, Anthony Lee Zhang, and Fudong Zhang.** 2025. "Land (mis)allocation and local public finance in China." Working paper.

- Huang, Yi, Marco Pagano, and Ugo Panizza.** 2020. "Local crowding-out in China." *The Journal of Finance*, 75(6): 2855–2898.
- Huet-Vaughn, Emiliano.** 2019. "Road spending, vote preference, and local partisanship." *American Economic Journal: Economic Policy*, 11(3): 156–186.
- Hu, Jiayin, Songrui Liu, Yang Yao, and Zhu Zong.** 2025. "Government deleveraging and the reverse crowding-in effect: Evidence from subnational debt and government contractors." *Management Science*.
- IMF.** 2024. "People's Republic of China: 2024 Article IV Consultation." International Monetary Fund Country Report No. 24/258.
- IMF.** 2025. "Fiscal Monitor: Fiscal Policy under Uncertainty." International Monetary Fund, Washington, DC.
- Janas, Marius.** 2026. "Public goods under financial distress." *Journal of Financial Economics*.
- Jia, Ruixue, Masayuki Kudamatsu, and David Seim.** 2015. "Political selection in China: The complementary roles of connections and performance." *Journal of the European Economic Association*, 13(4): 631–668.
- Koh, Yumi, Jing Li, and Jianhuan Xu.** 2025. "Subway, collaborative matching, and innovation." *The Review of Economics and Statistics*, 107(2): 476–493.
- Kose, M. Ayhan, Franziska Ohnsorge, Carmen M. Reinhart, and Kenneth S. Rogoff.** 2022. "The aftermath of debt surges." *Annual Review of Economics*, 14: 637–663.
- Lehne, Jonathan, Jacob N. Shapiro, and Oliver Vanden Eynde.** 2018. "Building connections: Political corruption and road construction in India." *Journal of Development Economics*, 131: 62–78.
- Lei, Zhenhuan, and Junlong Aaron Zhou.** 2022. "Private returns to public investment: Political career incentives and infrastructure investment in China." *The Journal of Politics*, 84(1): 455–469.
- Li, Hongbin, and Li-An Zhou.** 2005. "Political turnover and economic performance: the incentive role of personnel control in China." *Journal of Public Economics*, 89(9-10): 1743–1762.
- Li, Shanjun, Yanyan Liu, Avralt-Od Purevjav, and Lin Yang.** 2019. "Does subway expansion improve air quality?" *Journal of Environmental Economics and Management*, 96: 213–235.
- Mian, Atif, Ludwig Straub, and Amir Sufi.** 2025. "A goldilocks theory of fiscal deficits." *American Economic Review*, 115(12): 4253–4291.
- Mitchener, Kris James, and Christoph Trebesch.** 2023. "Sovereign debt in the twenty-first century." *Journal of Economic Literature*, 61(2): 565–623.

- Novy-Marx, Robert, and Joshua D. Rauh.** 2012. "Fiscal imbalances and borrowing costs: Evidence from state investment losses." *American Economic Journal: Economic Policy*, 4(2): 182–213.
- Painter, Marcus.** 2020. "An inconvenient cost: The effects of climate change on municipal bonds." *Journal of Financial Economics*, 135(2): 468–482.
- Qian, Shuoge, Hong Ru, and Wei Xiong.** 2025. "State versus market: China's infrastructure investment." NBER Working Paper.
- Reinhart, Carmen M., Vincent R. Reinhart, and Kenneth S. Rogoff.** 2012. "Public debt overhangs: Advanced-economy episodes since 1800." *Journal of Economic Perspectives*, 26(3): 69–86.
- Rogoff, Kenneth S., and Yuanchen Yang.** 2024. "A tale of tier 3 cities." *Journal of International Economics*, 152: 103989.
- Romer, Christina D., and David H. Romer.** 2019. "Fiscal space and the aftermath of financial crises: How it matters and why." *Brookings Papers on Economic Activity*, 2019(1): 239–313.
- Schwert, Michael.** 2017. "Municipal bond liquidity and default risk." *The Journal of Finance*, 72(4): 1683–1722.
- Severen, Christopher.** 2023. "Commuting, labor, and housing market effects of mass transportation: Welfare and identification." *Review of Economics and Statistics*, 105(5): 1073–1091.
- Song, Zheng Michael, and Wei Xiong.** 2026. "The Mandarin Model of Growth." Working paper.
- Sun, Liyang, and Sarah Abraham.** 2021. "Estimating dynamic treatment effects in event studies with heterogeneous treatment effects." *Journal of Econometrics*, 225(2): 175–199.
- Wang, Zhi, Qinghua Zhang, and Li-An Zhou.** 2020. "Career Incentives of City Leaders and Urban Spatial Expansion in China." *Review of Economics and Statistics*, 102(5): 897–911.
- Xiong, Wei.** 2025. "Derisking real estate in China's hybrid economy." Working paper.
- Xu, Chenggang.** 2011. "The Fundamental Institutions of China's Reforms and Development." *Journal of Economic Literature*, 1076–1151.
- Yared, Pierre.** 2019. "Rising government debt: Causes and solutions for a decades-old trend." *Journal of Economic Perspectives*, 33(2): 115–140.
- Zheng, Siqi, Yangfei Xu, Xiaonan Zhang, and Rui Wang.** 2016. "Transit development, consumer amenities and home values: Evidence from Beijing's subway neighborhoods." *Journal of Housing Economics*, 33: 22–33.
- Zhou, Feizhou.** 2012. *Profit as Profit: Fiscal Relations and Local Government Behavior (in Chinese)*. Shanghai: Shanghai Joint Publishing Company.

Panel A. Subway approval and the expansion of subway lines



Panel B. Share of subway in total municipal infrastructure investment

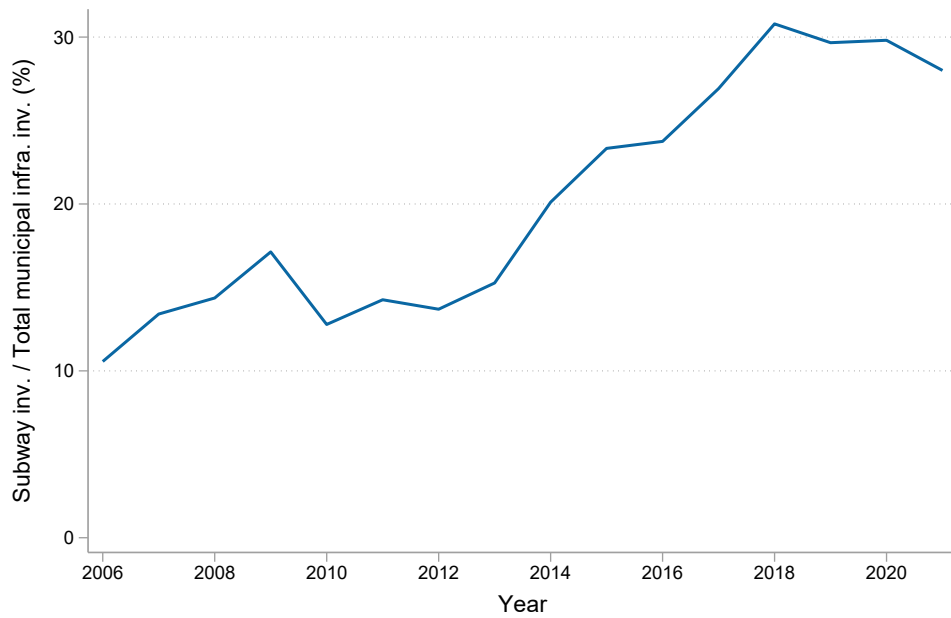


Figure 1. China's subway expansion: approvals and investment trends (2006–2022)

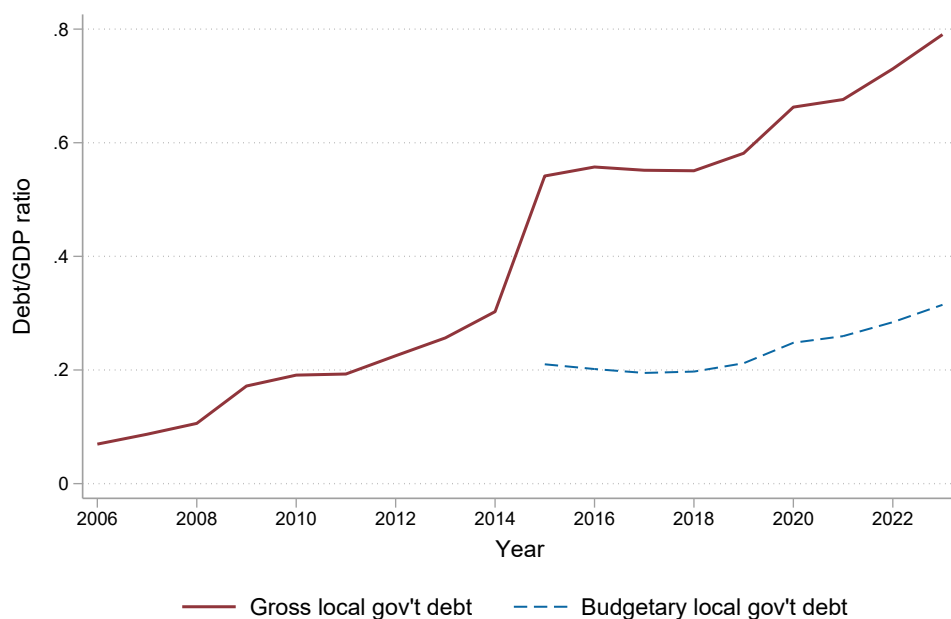


Figure 2. Local government debt-to-GDP ratio in China, 2006–2023

Notes: Gross local government debt includes LGFV interest-bearing liabilities and local government bonds. On-budget local government debt reflects government bond issuances recognized under the Budget Law introduced in 2015. Data are aggregated at the national level.

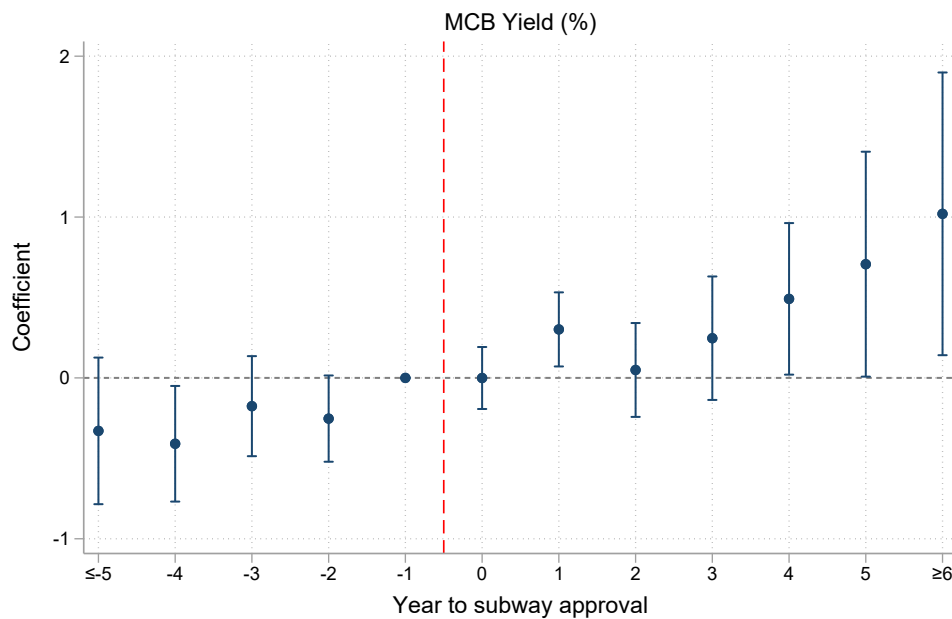


Figure 3. Event study: subway approval and municipal corporate bond yield

Notes: Event study results on the effects on MCB offering yield at bond level. Results are estimated following the approach by Sun and Abraham (2021) to account for heterogeneous treatment effects in staggered difference-in-differences estimation. The figure reports 95% confidence intervals, clustered at the city level.

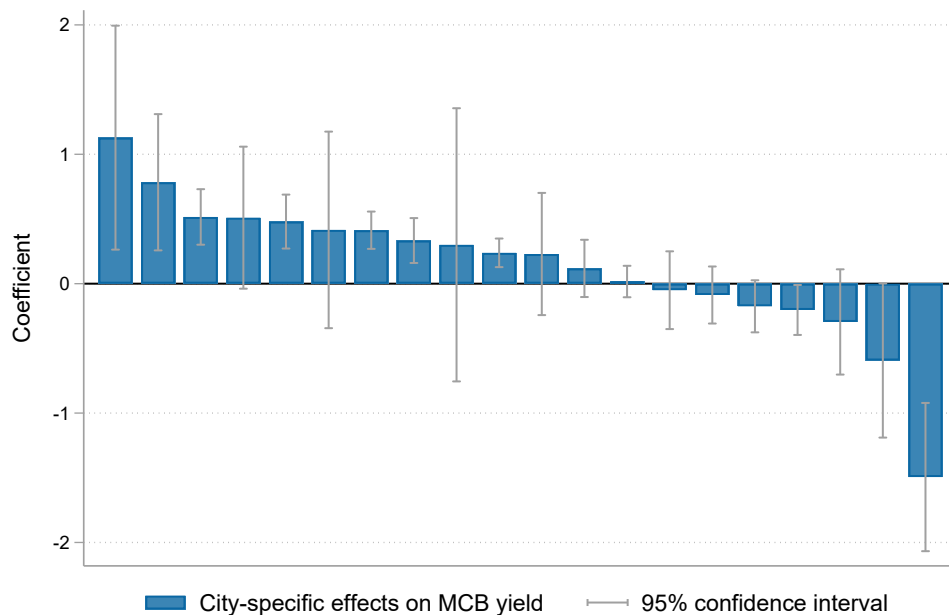


Figure 4. City-specific heterogeneous effects on MCB yield

Notes: This figure shows the city-specific effects on MCB yield. Bars indicate the estimated coefficients, and caps show the 95% confidence interval. **mark city names?**

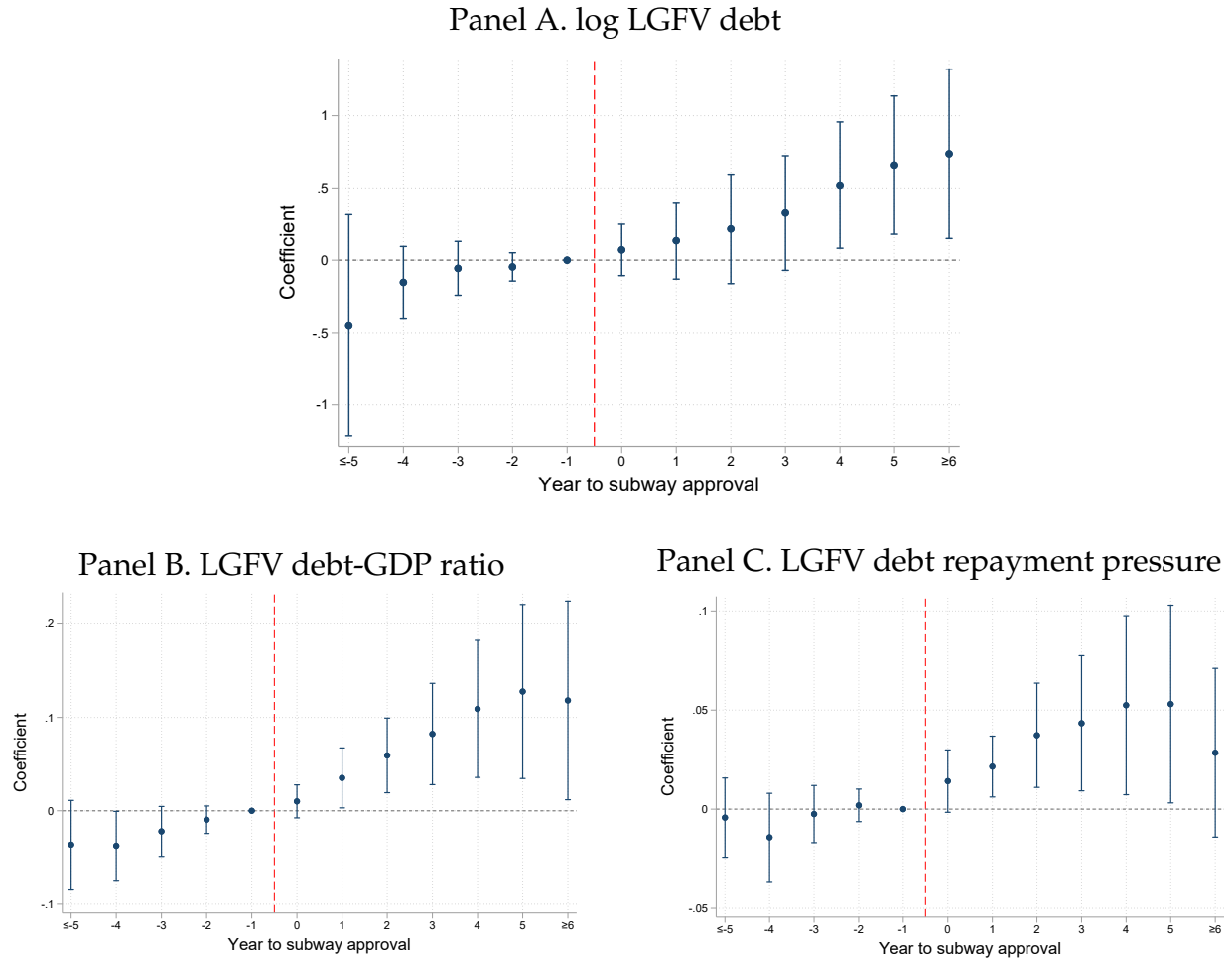


Figure 5. Event study: subway approval and local government debt level

Notes: We use the interest-bearing debt of Local Government Financing Vehicles (LGFVs) to measure local government debt burden. The dependent variables in Panels A to C are logged LGFV debt, LGFV debt-to-GDP ratio, and the amount of LGFV debt maturing in the subsequent year divided by GDP, respectively. Results are estimated following the approach by Sun and Abraham (2021) to account for heterogeneous treatment effects in staggered difference-in-differences estimation. The figure reports 95% confidence intervals, clustered at the city level.

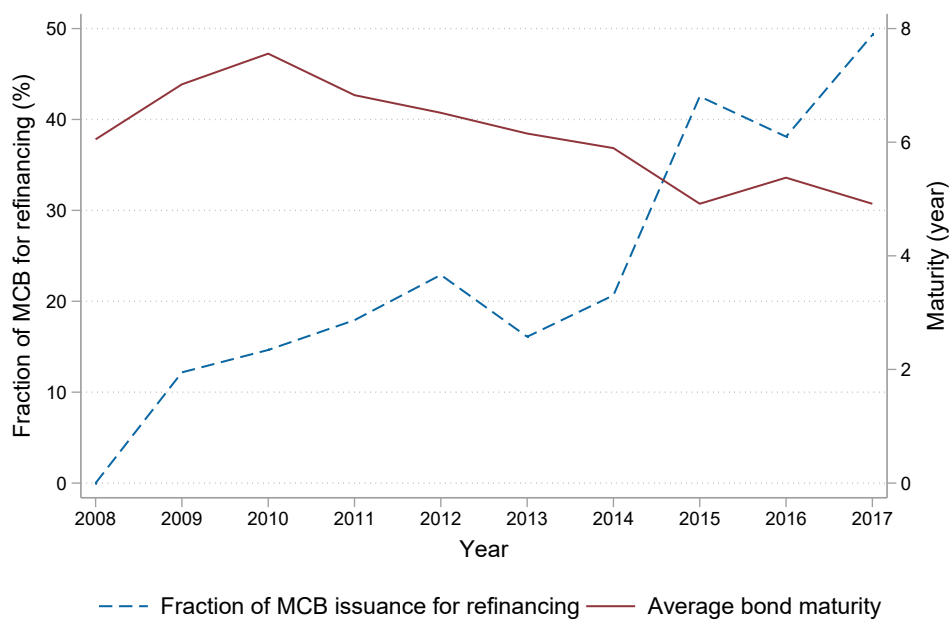


Figure 6. Share of MCB for repayment in total issuance and average maturity

Notes: Average bond maturity is weighted by bond issuance amount.

Table 1. Descriptive statistics of city characteristics by subway approval

	Mean			Difference	
	Control cities (1)	Control cities with subway plan (2)	Treated cities (3)	No control (4)	Control for Fiscal rev. (5)
Fiscal revenue (Bn RMB)	3.49	4.60	17.26	12.65*** (1.12)	
GDP (Bn RMB)	66.82	89.29	238.08	148.80*** (16.77)	-13.96 (12.45)
GDP per capita	22,131.70	22,520.12	53,917.07	31,396.96*** (5373.89)	2,592.77 (6,813.41)
Population (1,000)	3671.15	4606.19	5177.42	571.23 (491.29)	-619.37 (691.91)
Fiscal spending (Bn RMB)	8.19	10.12	21.28	11.16*** (1.24)	-0.08 (1.06)
Fiscal spending per capita	2,800.07	2,591.11	4,813.19	2,222.08*** (396.11)	589.45 (531.53)
LGFV Debt (Bn RMB)	5.11	8.11	37.66	29.55*** (4.05)	-0.96 (4.36)
LGFV Debt-GDP ratio	0.07	0.09	0.17	0.07*** (0.02)	0.04 (0.04)
Land sale revenue (Bn RMB)	1.92	1.18	4.67	3.49*** (0.58)	0.26 (0.74)
Land price (RMB/sqm)	1,524.55	1,043.78	1,640.86	597.08 (622.40)	105.45 (899.18)
Aging rate (%)	8.87	9.22	8.56	-0.67 (0.43)	-0.51 (0.62)
Number of cities	261	92	30		

Notes: Columns 1–3 shows the sample mean of predetermined city characteristics (measured in 2007). Column 1 reports the sample mean for control cities without any subway. Column 2 reports the refined control group cities that intended to build subway but had not obtained approval yet. Column 3 reports the mean for treated cities. Column 4 show the difference between treated cities and refined control cities (e.g., col. 2 vs. 3). Column 5 show the difference conditional on controlling for city's fiscal revenue (a key determinant of subway approval). Standard errors are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 2. Bond-level DID results: MCB offering yield

	Yield		
	(1)	(2)	(3)
Subway	0.230*** (0.061)	0.229*** (0.069)	0.202*** (0.076)
City FE	Y	Y	Y
Year FE	Y	Y	Y
Bond control	Y	Y	Y
City controls $\times$ Year FE		Y	Y
Subway plan sample			Y
Observations	8,486	8,430	5,681
Mean of D.V.	6.067	6.067	5.895

Notes: Bond controls include log issue size, time to maturity, indicator for guarantee, bond rating, and issuer rating. City controls include GDP, population, and fiscal revenues in 2008, interacted with year dummies. Robust standard errors in parentheses are clustered at city level. ***, **, * denote significance at 1%, 5%, 10% level.

Table 3. Heterogeneous effects on MCB yield

	Yield			
	(1)	(2)	(3)	(4)
Subway $\times$ Low ridership	0.345*** (0.085)	0.337*** (0.093)		
Subway $\times$ High ridership	0.040 (0.256)	0.087 (0.274)		
Subway $\times$ Low career incentive			0.148 (0.098)	0.132 (0.110)
Subway $\times$ High career incentive			0.286*** (0.074)	0.243*** (0.070)
City FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Bond control	Y	Y	Y	Y
City controls $\times$ Year FE		Y		Y
Subway plan sample	Y	Y	Y	Y
Observations	5,135	5,103	5,713	5,681
Mean of D.V.	5.895	5.895	5.895	5.895

Notes: High ridership is an indicator equal to one if a city's subway yearly ridership divided by the length of subway route is above the median of that in 2019. High career incentive is an indicator equal to one if the city party secretary's career incentive is above the median incentive for at the time of subway being approved. City controls include GDP, population, and fiscal revenues in 2008, interacted with year dummies. Robust standard errors in parentheses are clustered at city level. ***, **, * denote significance at 1%, 5%, 10% level.

Table 4. Subway and debt-financed investment

	log Infra. investment			Δ LGVFV debt
	All	From debt	From fiscal budget	
	(1)	(2)	(3)	(4)
Subway	0.411*** (0.136)	0.676** (0.321)	0.579** (0.263)	7.974** (3.327)
City FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
City FE $\times$ Year FE	Y	Y	Y	Y
City controls $\times$ Year FE	Y	Y	Y	Y
Observations	1,175	797	1,093	1,086
Mean of D.V.	3.130	1.384	1.565	7.964

Notes: City controls include GDP, population, and fiscal revenues in 2008, interacted with year dummies. Dependent variable in column 4 is calculated as the change in values of net increase in LGVFV debt (billions of RMB). Robust standard errors in parentheses are clustered at city level. ***, **, * denote significance at 1%, 5%, 10% level.

Table 5. Effects on fiscal revenue and residential land market

	log Tax revenue	log Land sale revenue	log Land sale area	log land price
	(1)	(2)	(3)	(4)
Subway	0.043 (0.027)	0.065 (0.105)	0.072 (0.107)	-0.016 (0.096)
City FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
City controls $\times$ Year FE	Y	Y	Y	Y
Subway plan cities	Y	Y	Y	Y
Observations	1,207	1,201	1,202	1,201

Notes: Dependent variable in column 4 is city-level area-weighted average price for residential land. City controls include GDP, population, and fiscal revenues in 2008, interacted with year dummies. Robust standard errors in parentheses are clustered at city level. ***, **, * denote significance at 1%, 5%, 10% level.

Table 6. Accumulating local government debt burden

	log Debt		Debt-GDP ratio		Debt repayment pressure	
	(1)	(2)	(3)	(4)	(5)	(6)
Subway	0.250** (0.108)	0.260** (0.107)	0.067*** (0.022)	0.070*** (0.021)	0.032* (0.017)	0.032* (0.017)
Debt swap		-0.405 (0.263)		-0.123** (0.049)		-0.010 (0.022)
City FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y
City controls $\times$ Year FE	Y	Y	Y	Y	Y	Y
Subway plan cities	Y	Y	Y	Y	Y	Y
Observations	1,197	1,194	1,208	1,198	1,210	1,200
Mean of D.V.	2.057	2.050	0.133	0.133	0.026	0.026

Notes: Debt repayment pressure is the amount of maturing LGVFV debt in the next year, divided by city GDP. Debt swap is the amount of special swap local governmental bond, scaled by the LGVFV outstanding debt in 2014, aggregated at the provincial level. City controls include GDP, population, and fiscal revenues in 2008, interacted with year dummies. Robust standard errors in parentheses are clustered at city level. ***, **, * denote significance at 1%, 5%, 10% level.

Table 7. Bond term structure and local debt burden

Bond sample	Yield			
	Short term (1)	Long term (2)	Short term (3)	Long term (4)
Subway	0.340* (0.157)	-0.086 (0.090)		
Subway $\times$ High debt burden			0.428* (0.182)	-0.105 (0.088)
Subway $\times$ Low debt burden			0.008 (0.178)	-0.005 (0.238)
City FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
City controls $\times$ Year FE	Y	Y	Y	Y
Bond controls	Y	Y	Y	Y
Subway plan sample	Y	Y	Y	Y
Observations	3,259	2,411	3,259	2,411

Notes: Columns 1 and 3 use short term bonds and columns 2 and 4 use long term bonds. We split the sample by whether the bond maturity is above five years. High debt burden is an indicator equal to one if a city's LGFV debt-GDP ratio is in the top quartile in 2008. City controls include GDP, population, and fiscal revenues in 2008, interacted with year dummies. Bond controls include log issue size, years to maturity, indicator for whether the bond has any guarantee, bond rating, and issuer rating.

Table 8. Rollover burden: MCB issuance by usage type

Bond type	MCB Issuance			
	(0/1)	amount	(0/1)	amount
	refinance (1)	refinance (2)	operating (3)	operating (4)
Subway	0.175* (0.069)	2.255* (1.004)	-0.053 (0.089)	1.648 (1.269)
City FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
City controls $\times$ Year FE	Y	Y	Y	Y
Subway plan cities	Y	Y	Y	Y
Mean of D.V.	0.310	1.504	0.503	2.524
Observations	1,210	1,210	1,210	1,210

Notes: This table reports the DID results on issuance of municipal corporate bonds by usage type. Dependent variables in columns 1 and 3 is the indicators for MCB issuance for purpose of debt refinance and operation or investment, respectively. Dependent variables in columns 2 and 4 is the issuance amount in billion RMB. City controls include GDP, population, and fiscal revenues in 2008, interacted with year dummies. Robust standard errors in parentheses are clustered at city level. ***, **, * denote significance at 1%, 5%, 10% level.

Table 9. Impacts on fiscal spending and public goods

	log Health spending (1)	Student- teacher ratio (2)	Land supply share for public goods (3)
<i>Panel A: Average effect</i>			
Subway	-0.091* (0.037)	0.741* (0.391)	-0.018 (0.019)
Observations	1,103	1,080	1,132
<i>Panel B: Heterogeneous effect by subway efficiency</i>			
Subway $\times$ Low ridership	-0.121*** (0.035)	0.945* (0.537)	-0.024* (0.011)
Subway $\times$ High ridership	-0.034 (0.057)	0.362 (0.428)	-0.007 (0.044)
Observations	1,103	1,080	1,132
City FE	Y	Y	Y
Year FE	Y	Y	Y
City controls $\times$ Year FE	Y	Y	Y

Notes: City controls include GDP, population, and fiscal revenues in 2008, interacted with year dummies. Land supply share for public goods is calculated as the area of land zoned for public facility, education, health, green land, and civil infrastructure (net of transportation), divided by the total area of newly supplied urban construction land. Robust standard errors in parentheses are clustered at city level. ***, **, * denote significance at 1%, 5%, 10% level.

Appendix

A Additional figures and tables

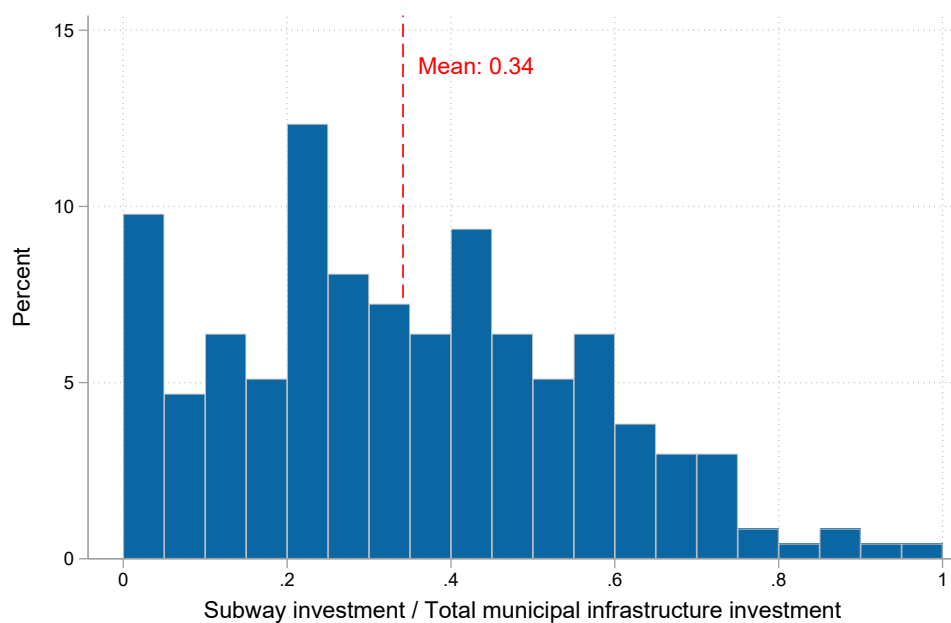


Figure A1. Share of subway investment in total municipal infrastructure investment in treated cities

Notes: This figure plots the distribution of investment in rail and transit as a percentage of total municipal infrastructure investment in cities after the approval of the subway project by NDRC.

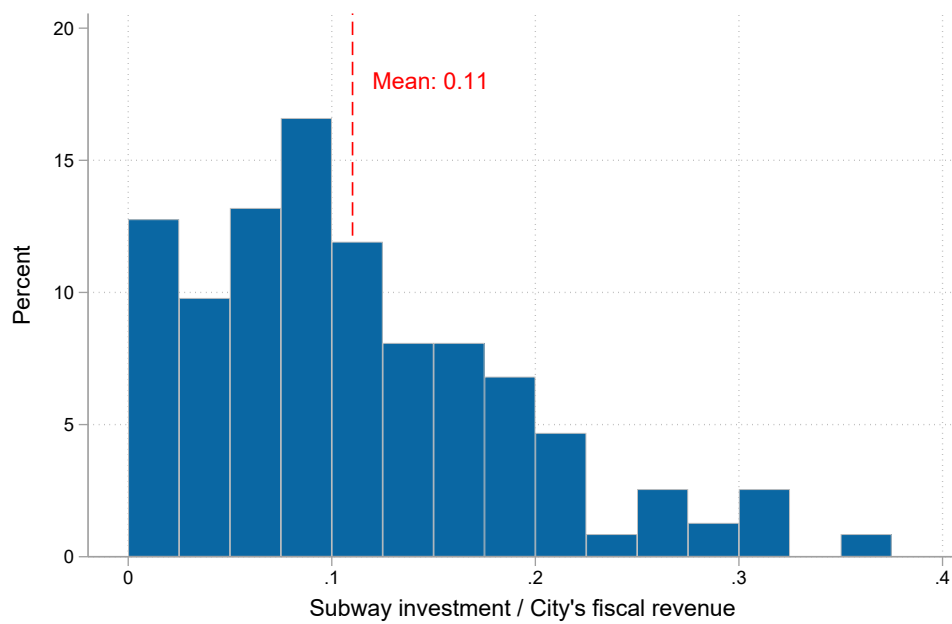


Figure A2. Ratio of subway investment/city's fiscal revenue in treated cities

Notes: This figure plots the distribution of investment in subway as a percentage of the city's general fiscal revenue in cities after the approval of the subway project by NDRC.

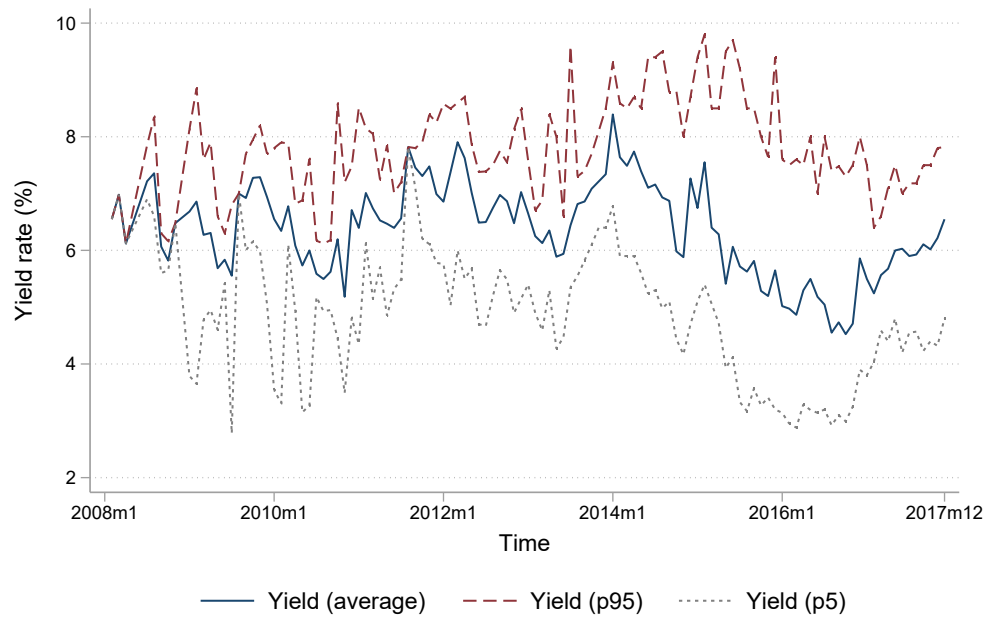


Figure A3. Municipal corporate bond yields over time: average, p5, and p95

Notes: This figure shows the monthly average MCB yields along with the 95th and 5th percentiles, weighted by bond issuance amount, in our analysis sample of 8,491 MCBs issued from 2008 to 2017.

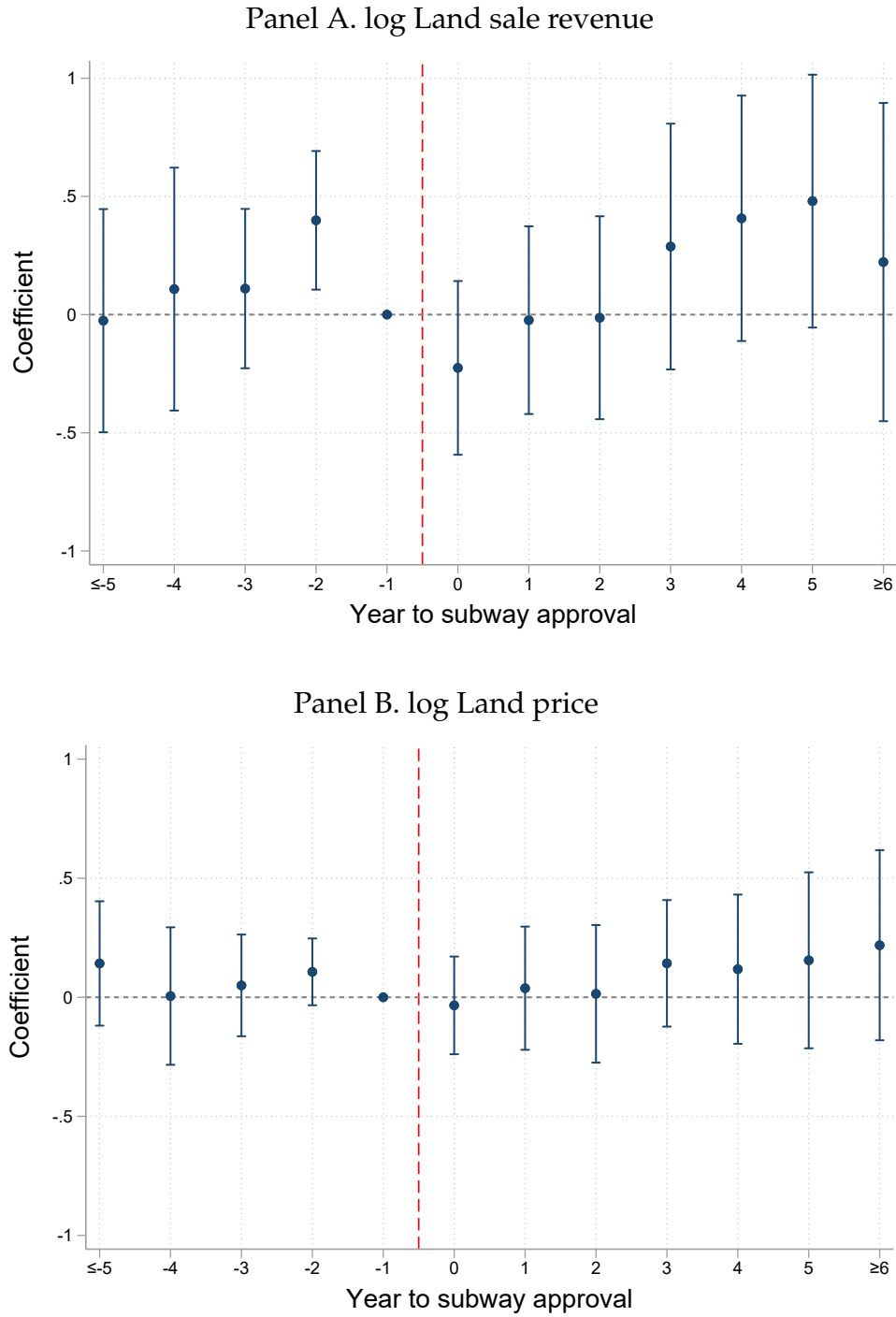


Figure A4. Event study: subway approval and residential land sale: revenue and price

Notes: This figure plots the event-study results of the effect of subway approval using city-year level data. The dependent variables in panels A and B are the logged land sale revenue and land price of residential land, respectively. Results are estimated following the approach by Sun and Abraham (2021) to account for heterogeneous treatment effects in staggered difference-in-differences estimation. The figure reports 95% confidence intervals, clustered at the city level.

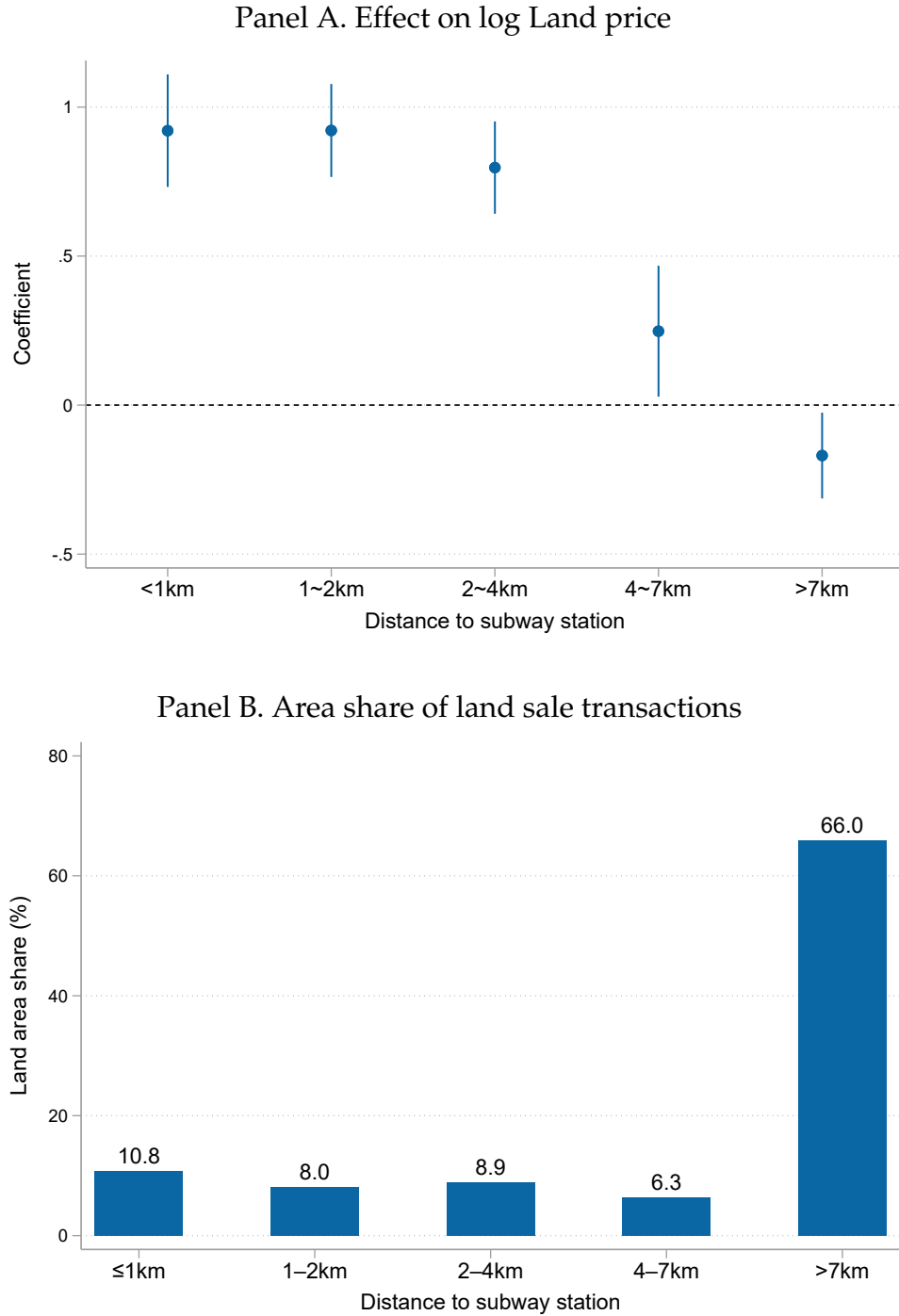


Figure A5. Heterogeneous effect on land price by distance to subway station

Notes: Panel A plots the effect on land price by distance of land parcel to subway station, using land transaction level data. The dependent variable is the log land price. Robust standard errors are clustered at the agent level. The figure reports 95% confidence intervals. Standard errors clustered at the city level. Panel B plots the area share of residential land sale transactions by distance to subway station in treated cities after the approval of subway projects by the NDRC.

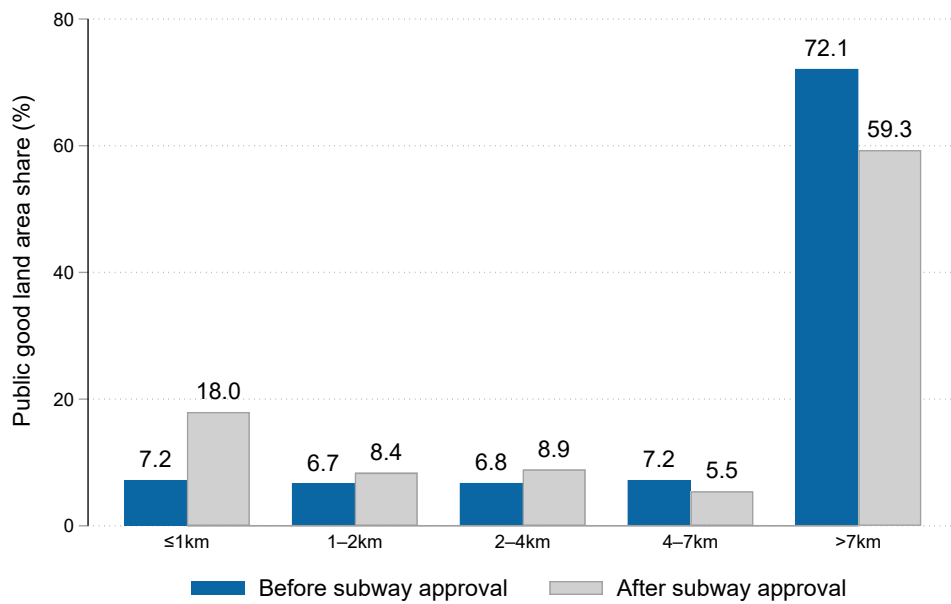


Figure A6. Area share of land supply for public goods

Notes: This figure plots the area share of land supply for public goods by distance to the nearest subway station in treated cities before and after the approval of the subway project by the NDRC. Land supply for public goods includes land zoned for public facilities, education, health, green land, and civil infrastructure (excluding roads and transportation).

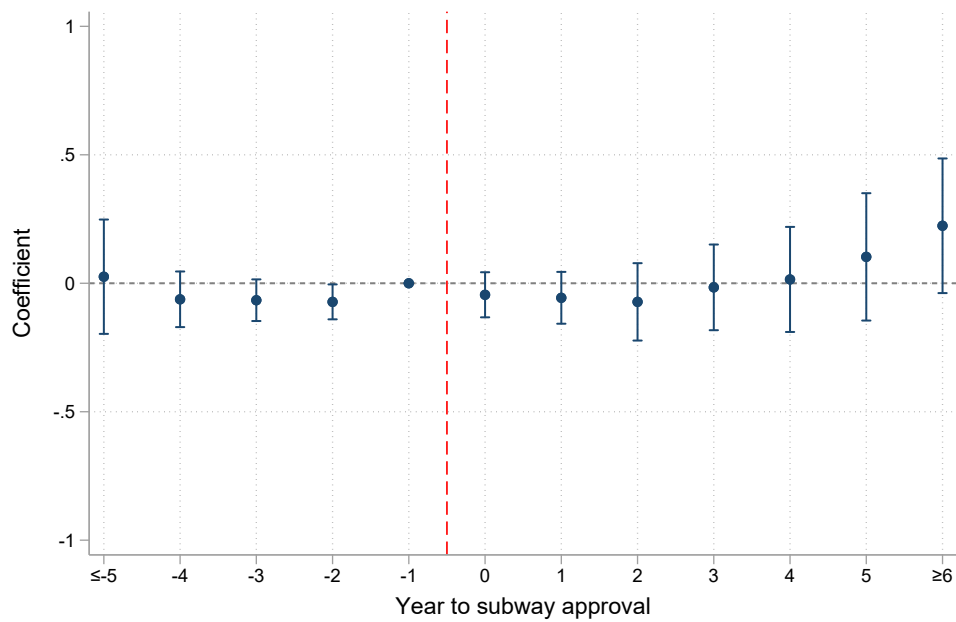


Figure A7. Event study: subway approval and real estate investment

Notes: This figure plots the event-study results of the effect of subway approval using city-year level data. The dependent variable is the logged real estate investment. Results are estimated following the approach by Sun and Abraham (2021) to account for heterogeneous treatment effects in staggered difference-in-differences estimation. The figure reports 95% confidence intervals, clustered at the city level.

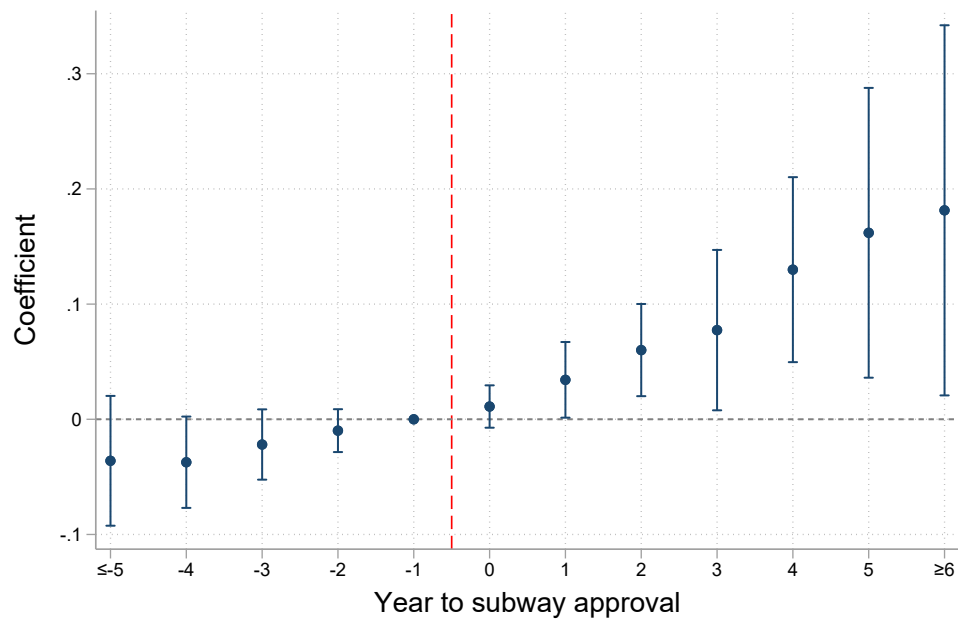


Figure A8. Event study: Subway approval and local government debt level, alternative measure

Notes: This figure plots the event-study results of the effect of subway approval using city-year level data. The dependent variable is gross local government debt-to-GDP ratio. Gross local government debt includes LGFV interest-bearing debt and budgetary local government bond after the introduction of new Budget Law in 2015. Results are estimated following the approach by Sun and Abraham (2021) to account for heterogeneous treatment effects in staggered difference-in-differences estimation. The figure reports 95% confidence intervals, clustered at the city level.

Table A1. Summary statistics of municipal corporate bonds (MCB)

	N	mean	S.D.	P5	P95
<i>Panel A: Full sample</i>					
Yield	8,491	6.07	1.44	3.64	8.30
Yield spread	8,491	2.79	1.18	1.00	4.85
Issue size (billion RMB)	8,491	0.97	0.57	0.27	2.00
Maturity (year)	8,491	5.18	2.36	0.74	7.00
Guarantee	8,491	0.19	0.40	0.00	1.00
Bond rated AA+ or AAA	8,491	0.25	0.43	0.00	1.00
Issuer rated AA+ or AAA	8,491	0.23	0.42	0.00	1.00
<i>Panel B: City with intended subway plan</i>					
Yield	5,716	5.90	1.42	3.53	8.12
Yield spread	5,716	2.65	1.15	0.93	4.70
Issue size (billion RMB)	5,716	0.99	0.61	0.25	2.00
Maturity (year)	5,716	4.98	2.43	0.74	7.00
Guarantee	5,716	0.18	0.38	0.00	1.00
Bond rated AA+ or AAA	5,716	0.27	0.44	0.00	1.00
Issuer rated AA+ or AAA	5,716	0.30	0.46	0.00	1.00

Notes: The table provides the summary statistics of MCBs issued from 2008 to 2017. Panel A reports the full sample including 30 cities as the treatment group that received subway plan approvals, and 256 cities as the never-treated group. Panel B restricts the control group sample to cities that have proposed subway plans but have not yet obtained approval.

Table A2. List of subway approvals

City	Approval year	Subway plan (in original Chinese)
Ningbo	2008	宁波市城市快速轨道交通建设规划（2006-2015 年）
Wuxi	2008	无锡市城市轨道交通近期建设规划（2008-2015 年）
Dongguan	2009	东莞市城市快速轨道交通建设规划（2009-2015 年）
Nanchang	2009	南昌市城市轨道交通近期建设规划（2009-2016 年）
Dalian	2009	大连市城市轨道交通建设规划（2009-2016 年）
Dalian	2009	大连市城市轨道交通建设规划（2009-2016 年）
Kunming	2009	昆明市城市快速轨道交通近期建设规划（2008-2016 年）
Fuzhou	2009	福州市城市快速轨道交通近期建设规划（2008-2016 年）
Zhengzhou	2009	郑州市城市快速轨道交通近期建设规划（2008-2015 年）
Zhengzhou	2009	郑州市城市快速轨道交通近期建设规划（2008-2015 年）
Changsha	2009	长沙市轨道交通第一轮建设规划（2008-2015 年）
Qingdao	2009	青岛市城市轨道交通近期建设规划（2009-2016 年）
Nanning	2010	南宁市城市轨道交通近期建设规划（2009-2015 年）
Hefei	2010	合肥市城市轨道交通近期建设规划（2009-2016 年）
Guiyang	2010	贵阳市城市轨道交通近期建设规划（2010-2020 年）
Changchun	2010	长春市城市快速轨道交通建设规划（2010 年-2016 年）
Xiamen	2011	厦门市城市轨道交通近期建设规划（2011-2020 年）
Foshan	2012	佛山市城市轨道交通近期建设规划（2011-2018 年）
Foshan	2012	佛山市城市轨道交通近期建设规划（2011-2018 年）
Ürümqi	2012	乌鲁木齐市城市轨道交通建设规划（2012-2019 年）
Lanzhou	2012	兰州市城市轨道交通近期建设规划（2011-2020 年）
Taiyuan	2012	太原市城市轨道交通近期建设规划（2012-2018 年）
Changzhou	2012	常州市城市轨道交通近期建设规划（2011-2018 年）
Shijiazhuang	2012	石家庄市城市轨道交通近期建设规划（2012-2020 年）
Wenzhou	2012	浙江省温州市域铁路建设规划（2012-2018 年）
Xuzhou	2013	徐州市城市轨道交通近期建设规划（2013-2020 年）
Nantong	2014	南通市城市轨道交通近期建设规划（2014-2020 年）
Hohhot	2015	呼和浩特市城市轨道交通近期建设规划（2015-2020 年）
Jinan	2015	济南市城市轨道交通近期建设规划（2015-2019 年）
Baotou	2016	包头市城市轨道交通第一期建设规划（2016-2022 年）
Baotou	2016	包头市城市轨道交通第一期建设规划（2016-2022 年）
Luoyang	2016	洛阳市城市轨道交通第一期建设规划（2016-2020 年）
Shaoxing	2016	绍兴市城市轨道交通第一期建设规划（2016-2021 年）
Wuhu	2016	芜湖市轨道交通建设规划（2016 年-2020 年）

Notes: This table lists the cities that received subway plan approvals along with the approval years and the original Chinese names of the subway construction plans.

Table A3. Determinants of subway approvals: Cox hazard analysis

	Subway approved (0/1)		
	(1)	(2)	(3)
log Fiscal revenue	4.103*** (0.844)	4.047** (1.623)	4.047** (1.623)
log GDP	-0.844 (0.870)	-0.518 (1.029)	-0.518 (1.029)
log Population	-0.043 (0.385)	0.849 (0.621)	0.849 (0.621)
log Fiscal spending		-0.429 (1.616)	-0.429 (1.616)
log LGV Debt		0.397 (0.482)	0.397 (0.482)
LGV Debt-GDP ratio		0.744 (3.137)	0.744 (3.137)
log land sale revenue		-0.548 (0.346)	-0.548 (0.346)
log land price		0.516 (0.368)	0.516 (0.368)
Aging rate (%)		-0.177 (0.114)	-0.177 (0.114)
Year FE			Y
Observations	2,729	2,606	2,606

Notes: This table presents the estimated coefficients of a Cox hazard model. The dependent variable is a dummy equal to one if a city received approval for subway construction in year t , and 0 otherwise. Once a city obtained approval, it is dropped from the sample. ***, **, * denote significance at 1%, 5%, 10% level.

Table A4. Effects on fiscal revenue per capita and population

	log Land sale revenue (1)	log Total fiscal revenue (2)	log Tax revenue (3)
Subway	0.052 (0.107)	0.013 (0.030)	0.031 (0.028)
City FE	Y	Y	Y
Year FE	Y	Y	Y
City controls $\times$ Year FE	Y	Y	Y
Subway plan cities	Y	Y	Y
Observations	1,210	1,207	1,201

Notes: Robust standard errors in parentheses are clustered at city level. ***, **, * denote significance at 1%, 5%, 10% level.

Table A5. Effect on land price: land parcel transaction level results

	log Land price		
	(1)	(2)	(3)
Subway	0.010 (0.073)	0.007 (0.079)	0.019 (0.076)
City FE		Y	Y
Year FE	Y	Y	Y
Subway plan cities	Y	Y	Y
City controls $\times$ Year FE		Y	Y
Land controls			Y
Observations	122,608	121,349	121,210

Notes: The unit of observation is the land parcel. The sample is restricted to residential land sale by auctions. City controls include GDP, population, and fiscal revenues in 2008, interacted with year dummies. Land controls including area and land grade. Robust standard errors in parentheses are clustered at city level. ***, **, * denote significance at 1%, 5%, 10% level.

Table A6. Local government debt burden, alternative measure

	LGVFV Debt/ total budget		Gross debt/ GDP		Gross debt/ total budget	
	(1)	(2)	(3)	(4)	(5)	(6)
Subway	0.385*** (0.137)	0.400*** (0.133)	0.069*** (0.026)	0.067*** (0.025)	0.395** (0.161)	0.392** (0.159)
Debt swap		-0.730*** (0.266)		0.082 (0.088)		0.153 (0.258)
City FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y
City controls $\times$ Year FE	Y	Y	Y	Y	Y	Y
Subway plan cities	Y	Y	Y	Y	Y	Y
Observations	831	831	1,139	1,130	1,877	1,877
Mean of D.V.	0.780	0.780	0.206	0.206	1.037	1.037

Notes: City's total fiscal budget is defined as the sum of general public budget revenues, fiscal transfers, and land sale revenues. Gross local government debt includes LGVFV interest-bearing debt and budgetary local government bond after the introduction of new Budget Law in 2015. City controls include GDP, population, and fiscal revenues in 2008, interacted with year dummies. Robust standard errors in parentheses are clustered at city level. ***, **, * denote significance at 1%, 5%, 10% level.

Table A7. City-year level outcomes: synthetic difference-in-difference

<i>Panel A: Fiscal outcomes</i>			
	log Tax revenue (1)	log Land sale revenue (2)	log Land price (3)
Subway	-0.022 (0.057)	-0.103 (0.203)	0.084 (0.127)
Observations	1,180	1,180	1,180
<i>Panel B: Debt outcomes</i>			
	Debt-GDP ratio (1)	Debt repayment pressure (2)	MCB Issuance Refinance (=1) (3)
Subway	0.056*** (0.015)	0.034*** (0.004)	0.433*** (0.066)
Observations	1,190	1,200	1,200
City FE	Y	Y	Y
Year FE	Y	Y	Y
Year FE $\times$ Subway plan cities	Y	Y	Y

Notes: This table reports results of major city-year level outcomes using synthetic difference-in-difference estimator by Arkhangelsky et al. (2021). ***, **, * denote significance at 1%, 5%, 10% level.

Table A8. City-year level outcomes: DID estimator by Callaway and Sant'Anna (2021)

<i>Panel A: Fiscal outcomes</i>			
	log Tax revenue (1)	log Land sale revenue (2)	log Land price (3)
Subway	0.057 (0.046)	0.083 (0.180)	0.059 (0.118)
Observations	1,197	1,191	1,191
<i>Panel B: Debt outcomes</i>			
	Debt-GDP ratio (1)	Debt repayment pressure (2)	MCB Issuance Refinance (=1) (3)
Subway	0.079*** (0.024)	0.033** (0.015)	0.325*** (0.099)
Observations	1,198	1,200	1,200
City FE	Y	Y	Y
Year FE	Y	Y	Y
City controls $\times$ Year FE	Y	Y	Y
Subway plan cities	Y	Y	Y

Notes: This table reports results of major city-year level outcomes using another staggered DID estimator following Callaway and Sant'Anna (2021). ***, **, * denote significance at 1%, 5%, 10% level.

B Robustness checks

Table B1. Bond level DID results: interaction-weighted estimator by Sun and Abraham (2021)

	Yield	
	(1)	(2)
Subway	0.396*** (0.150)	0.402** (0.193)
City FE	Y	Y
Year FE	Y	Y
Bond controls	Y	Y
City controls $\times$ Year FE		Y
Observations	8,430	8,430
Mean of D.V.	2.794	2.794

Notes: Bond controls include log issue size, time to maturity, indicator for guarantee, bond rating, and issuer rating. City controls include GDP, population, and fiscal revenues in 2008, interacted with year dummies. Robust standard errors in parentheses are clustered at city level. ***, **, * denote significance at 1%, 5%, 10% level.

Table B2. Bond-year level DID results: spread

	Spread	
	(1)	(2)
Subway	0.136** (0.057)	0.145** (0.071)
City FE	Y	Y
Year FE	Y	Y
Bond controls	Y	Y
City controls $\times$ Year FE		Y
Observations	8,486	8,430
Mean of D.V.	2.794	2.794

Notes: Spread is calculated as the difference between the MCB yield and treasury bond yield on the same day and of the same maturity. Bond controls include log issue size, time to maturity, indicator for guarantee, bond rating, and issuer rating. City controls include GDP, population, and fiscal revenues in 2008, interacted with year dummies. Robust standard errors in parentheses are clustered at city level. ***, **, * denote significance at 1%, 5%, 10% level.

Table B3. Bond-year level DID results: secondary yield and spread using bond transaction data

	Sec. yield		Sec. spread	
	(1)	(2)	(3)	(4)
Subway	0.271*** (0.102)	0.228** (0.103)	0.244** (0.100)	0.205** (0.101)
City FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Bond controls	Y	Y	Y	Y
City controls $\times$ Year FE		Y		Y
Observations	17,294	17,183	17,290	17,179
Mean of D.V.	5.36	5.36	2.07	2.07

Notes: Spread is calculated as the difference between the MCB yield and treasury bond yield on the same day and of the same maturity. Bond controls include log issue size, time to maturity, indicator for guarantee, bond rating, and issuer rating. City controls include GDP, population, and fiscal revenues in 2008, interacted with year dummies. Robust standard errors in parentheses are clustered at city level. ***, **, * denote significance at 1%, 5%, 10% level.

Table B4. Bond-level DID results: controlling for debt swap program

	Yield		
	(1)	(2)	(3)
Subway	0.233*** (0.057)	0.227*** (0.068)	0.205*** (0.075)
Debt swap	0.269 (0.181)	0.246 (0.187)	0.387 (0.246)
City FE	Y	Y	Y
Year FE	Y	Y	Y
Bond control	Y	Y	Y
City controls $\times$ Year FE		Y	Y
Subway plan sample			Y
Observations	8,486	8,430	5,681
Mean of D.V.	6.067	6.067	5.895

Notes: City controls include GDP, population, and fiscal revenues in 2008, interacted with year dummies. Debt swap is the amount of special swap local governmental bond, scaled by the LGVFFV outstanding debt in 2014, aggregated at the provincial level. Robust standard errors in parentheses are clustered at city level. ***, **, * denote significance at 1%, 5%, 10% level.